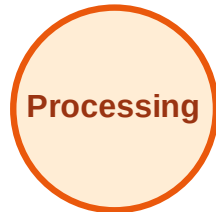


# BFS Traversal

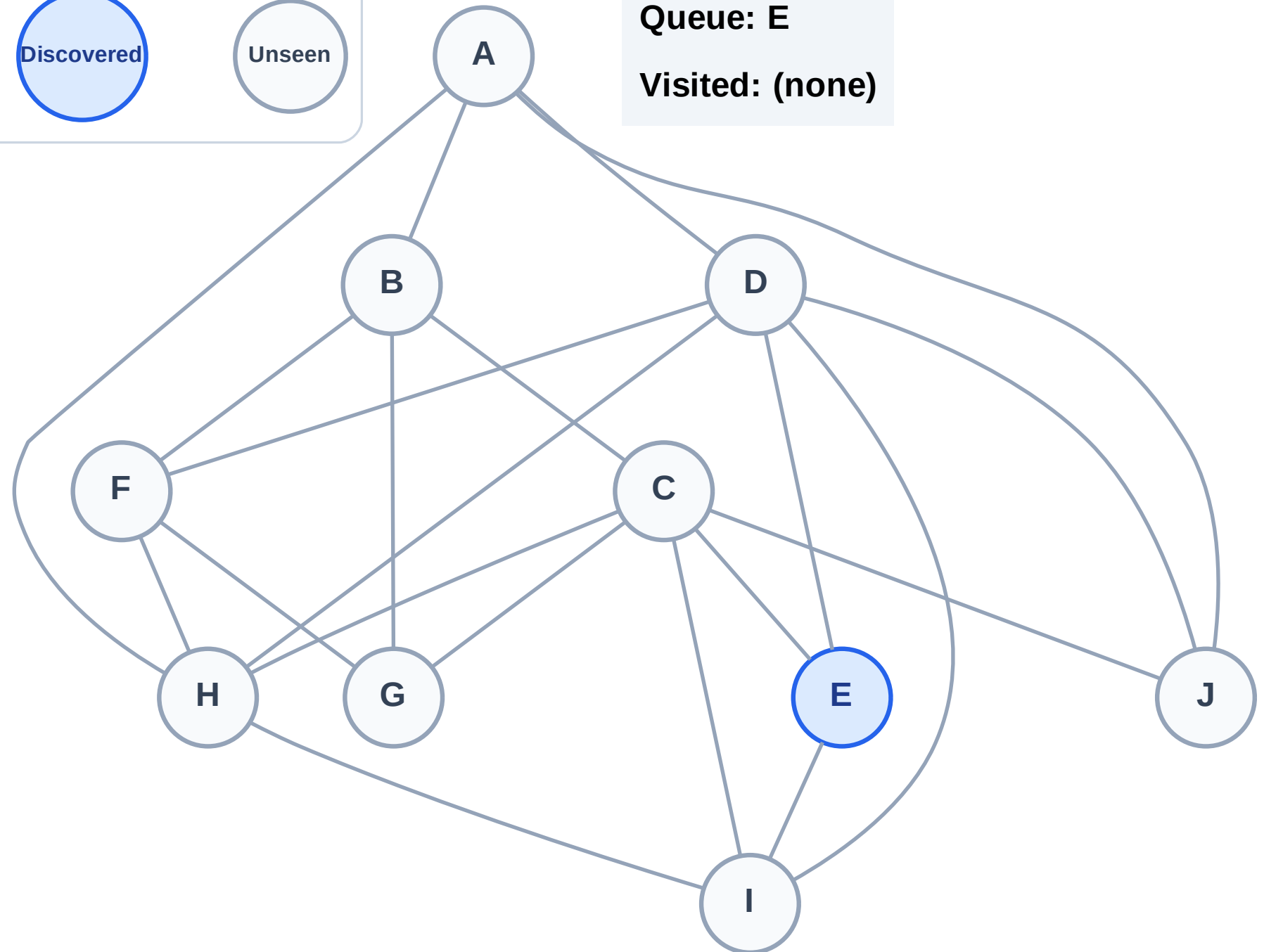
Step 1: Start at E

## Legend



Queue: E

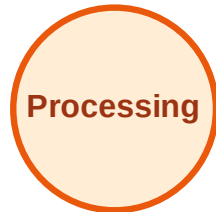
Visited: (none)



# BFS Traversal

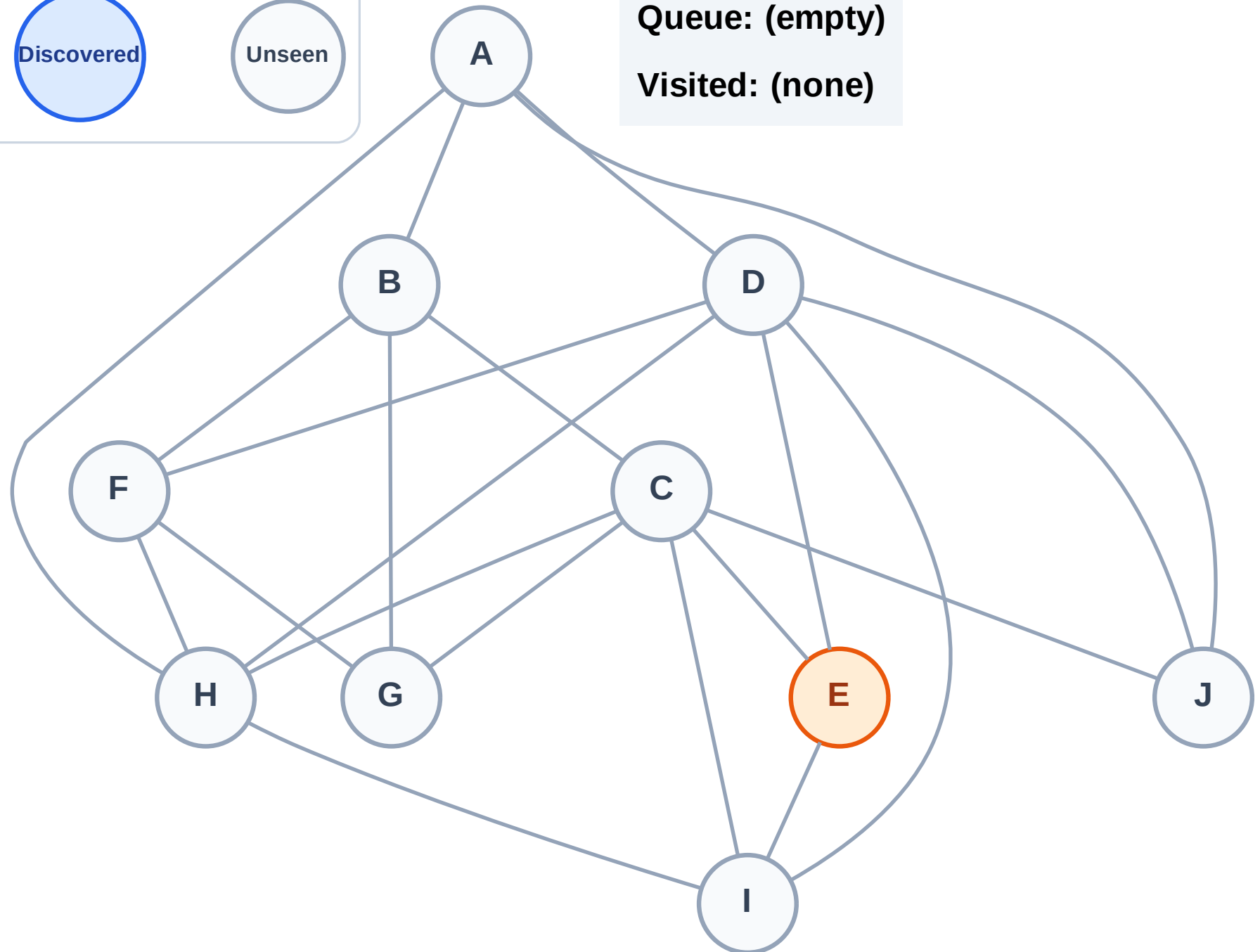
Step 2: Process node E

## Legend



Queue: (empty)

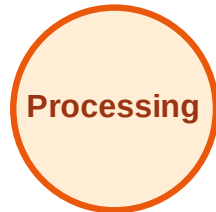
Visited: (none)



# BFS Traversal

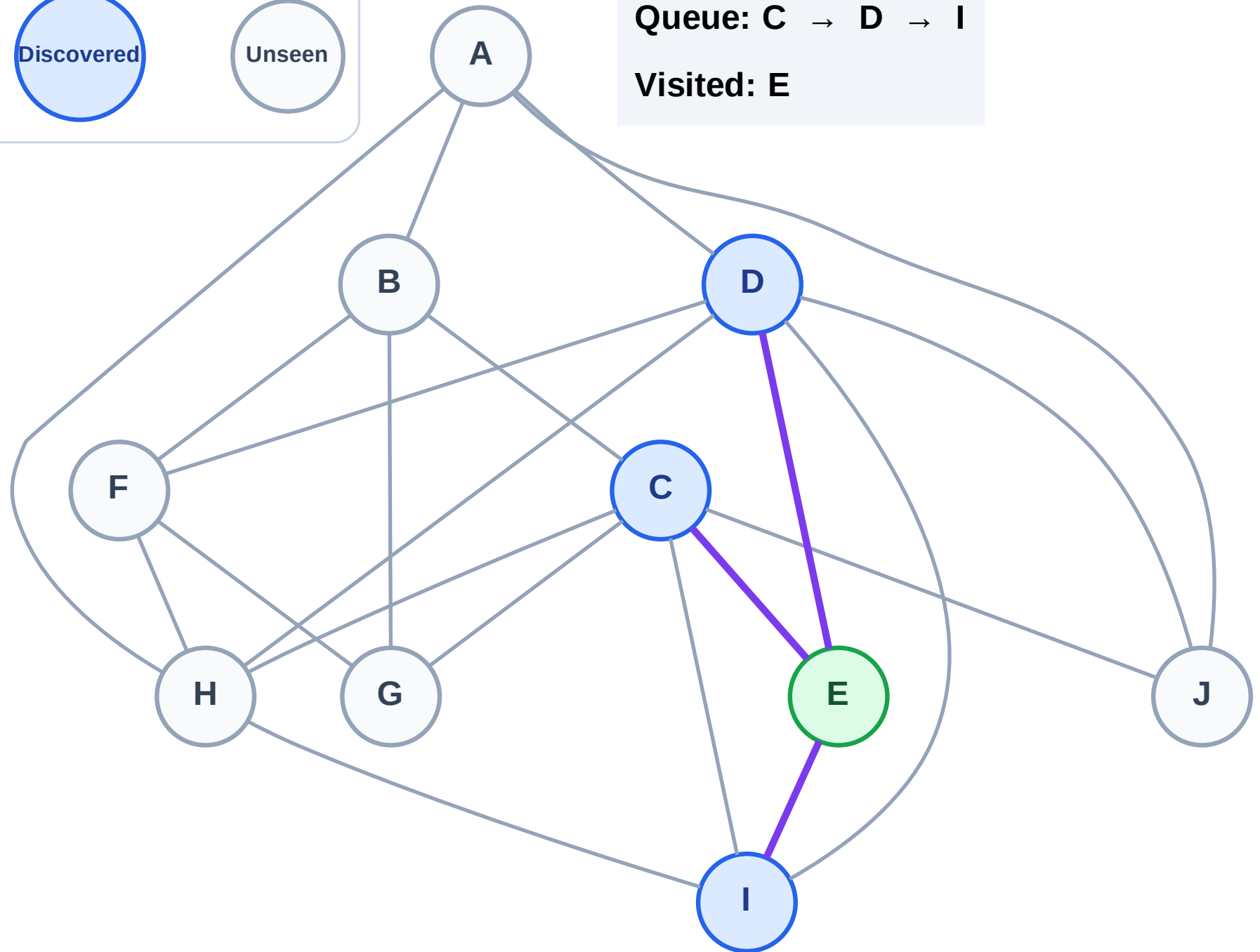
Step 3: Discover C, D, I from E

## Legend



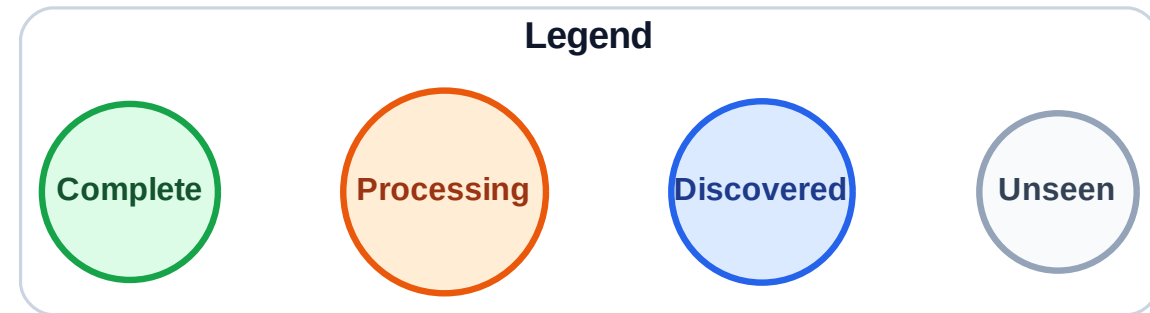
Queue: C → D → I

Visited: E



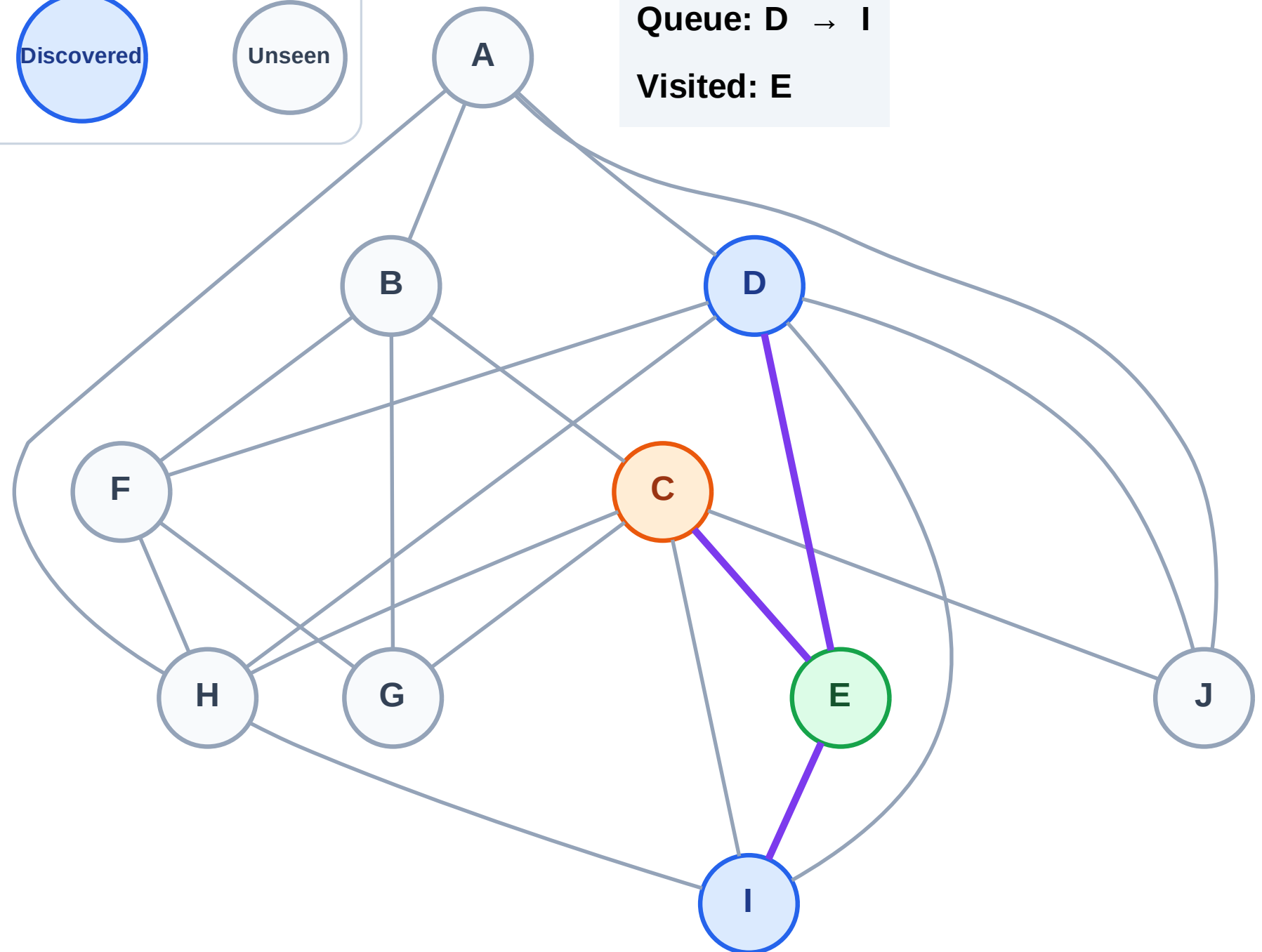
# BFS Traversal

Step 4: Process node C



Queue: D → I

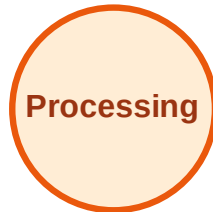
Visited: E



# BFS Traversal

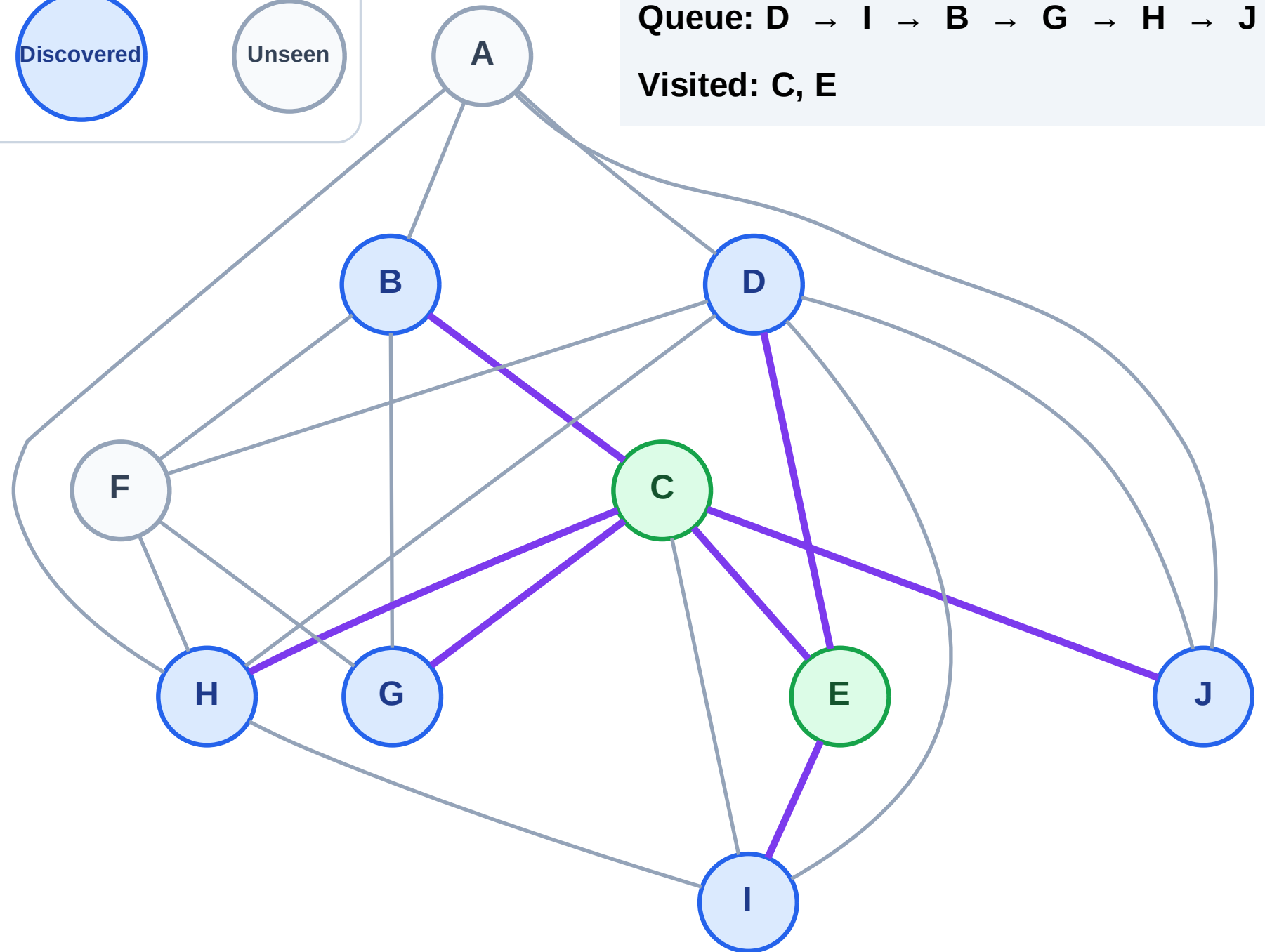
Step 5: Discover B, G, H, J from C

## Legend



Queue: D → I → B → G → H → J

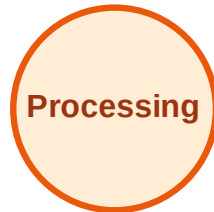
Visited: C, E



# BFS Traversal

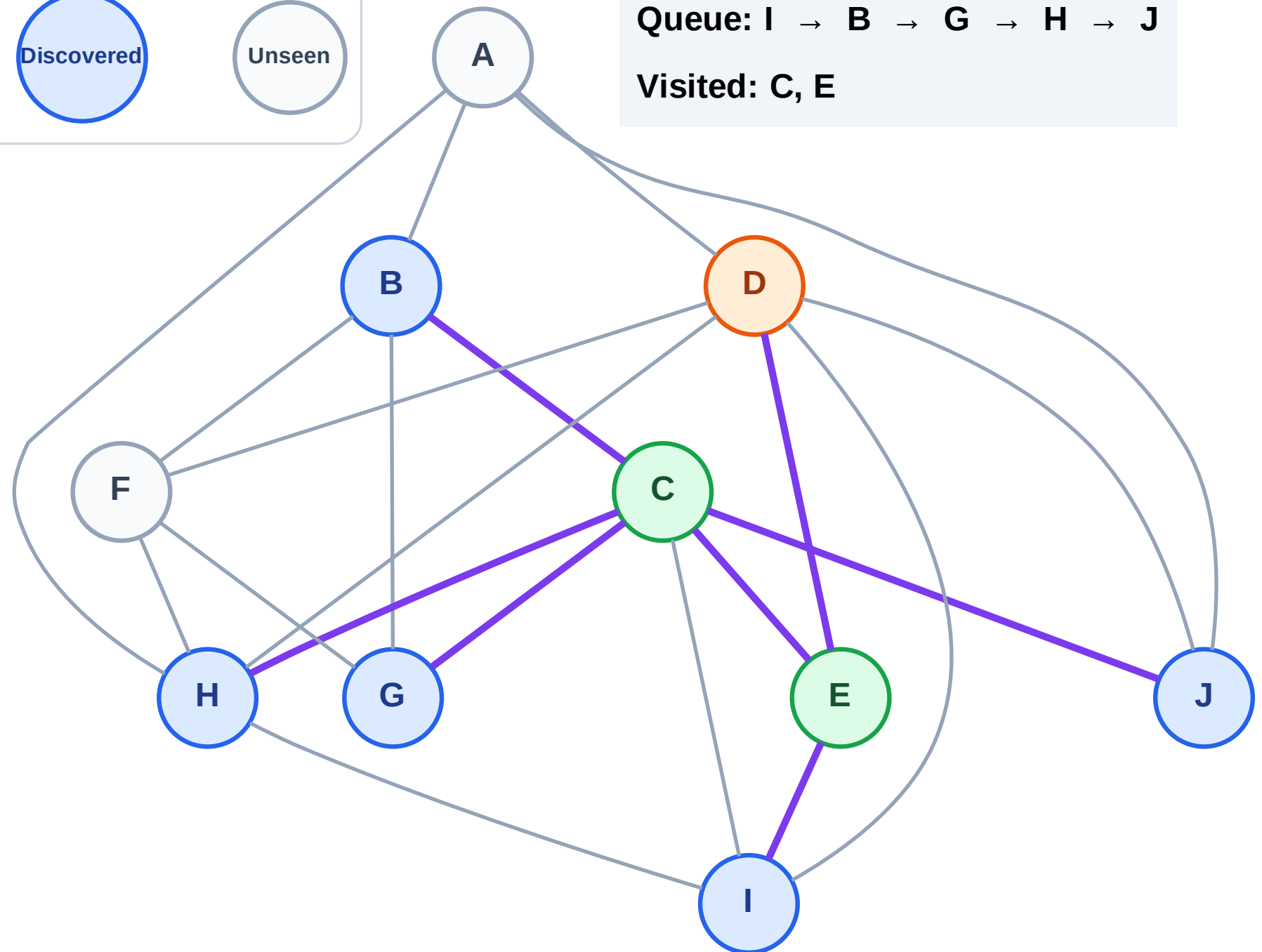
Step 6: Process node D

## Legend



Queue: I → B → G → H → J

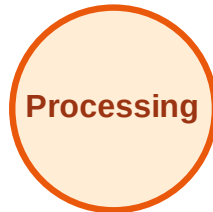
Visited: C, E



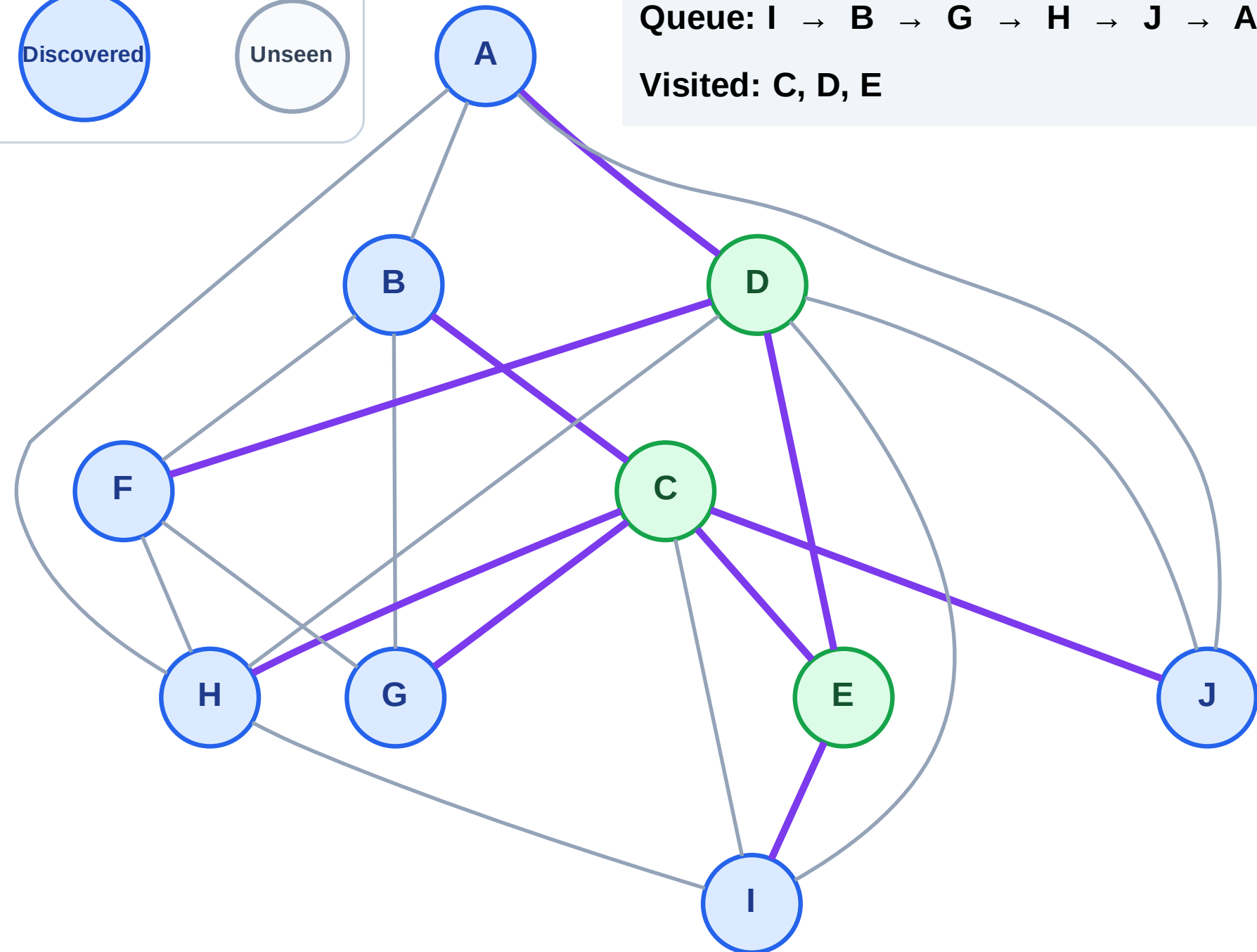
# BFS Traversal

Step 7: Discover A, F from D

## Legend



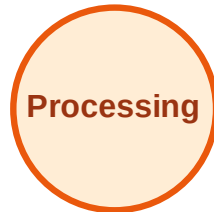
Queue: I → B → G → H → J → A → F  
Visited: C, D, E



# BFS Traversal

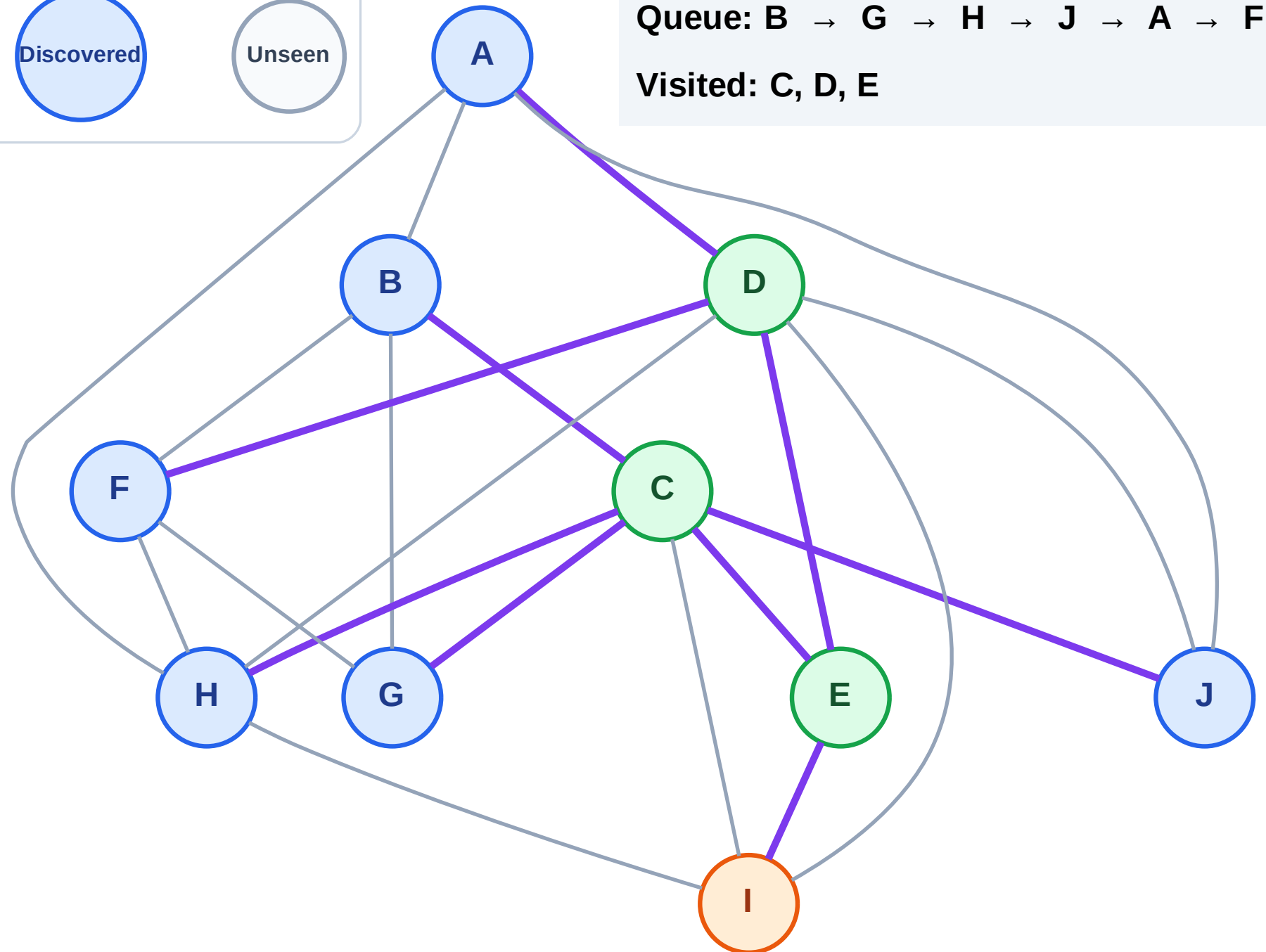
Step 8: Process node I

## Legend



Queue: B → G → H → J → A → F

Visited: C, D, E





# BFS Traversal

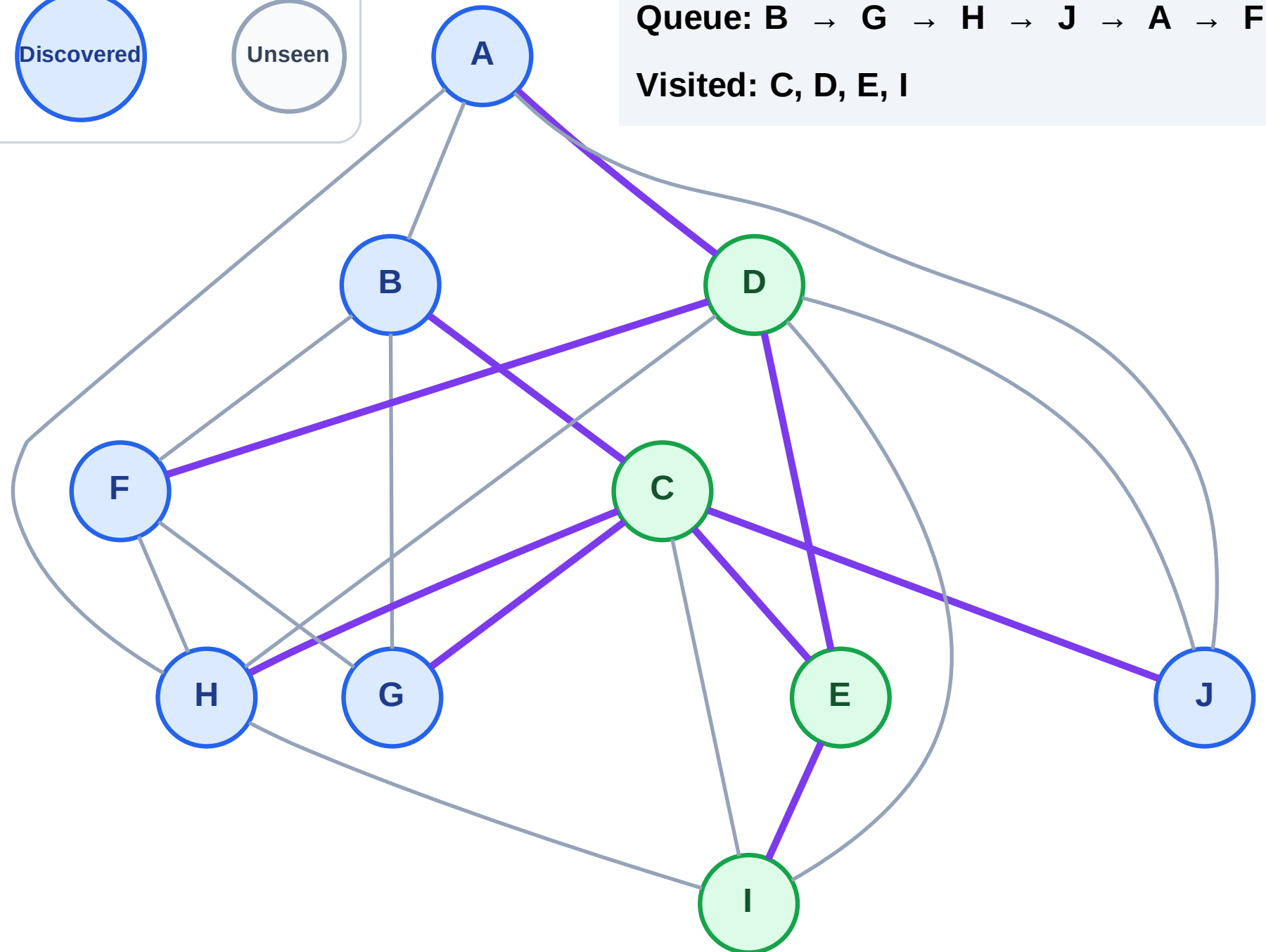
Step 9: No new neighbors from I

## Legend



Queue: B → G → H → J → A → F

Visited: C, D, E, I



# BFS Traversal

Step 10: Process node B

## Legend

Complete

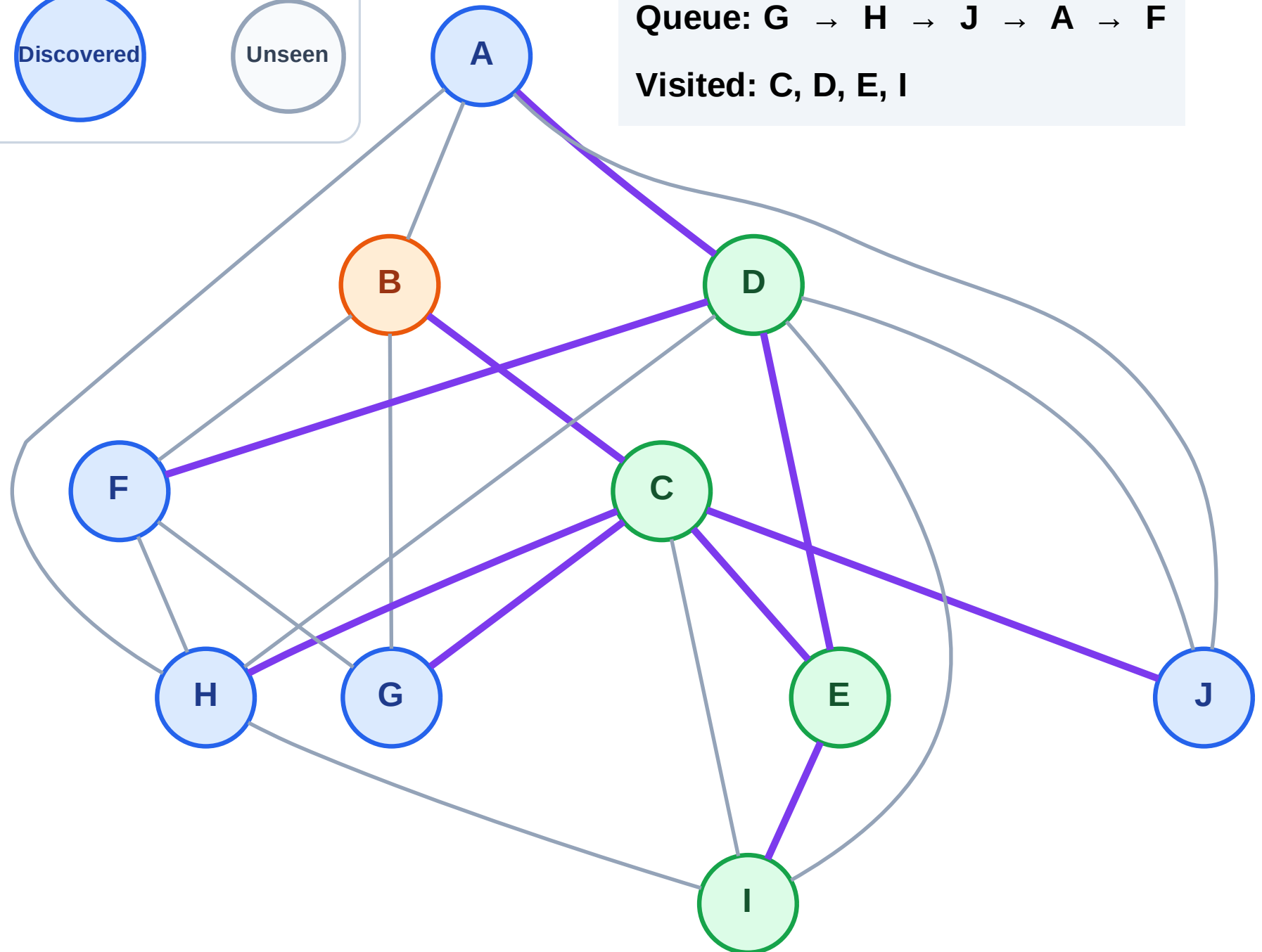
Processing

Discovered

Unseen

Queue: G → H → J → A → F

Visited: C, D, E, I



# BFS Traversal

Step 11: No new neighbors from B

## Legend

Complete

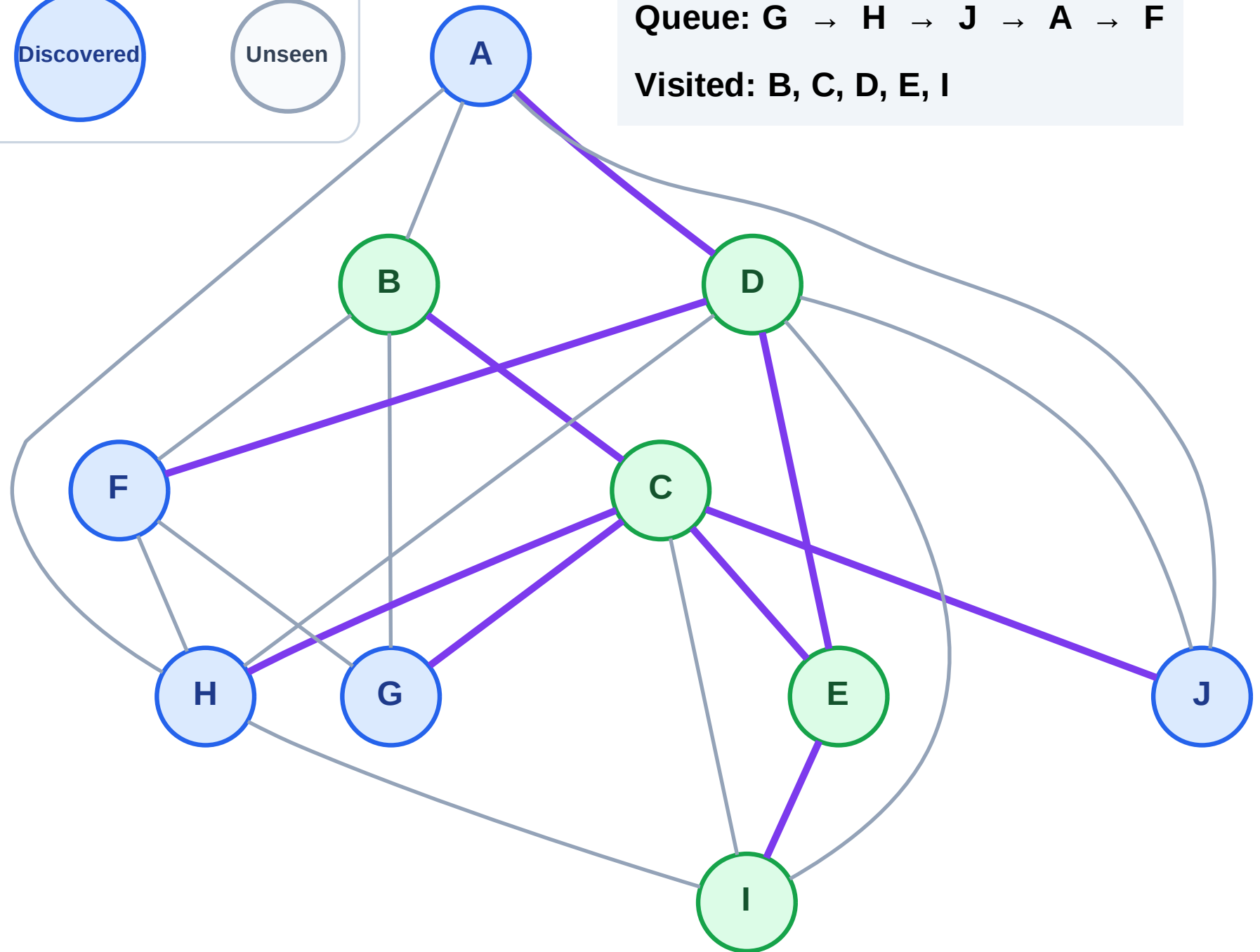
Processing

Discovered

Unseen

Queue: G → H → J → A → F

Visited: B, C, D, E, I



# BFS Traversal

Step 12: Process node G

## Legend

Complete

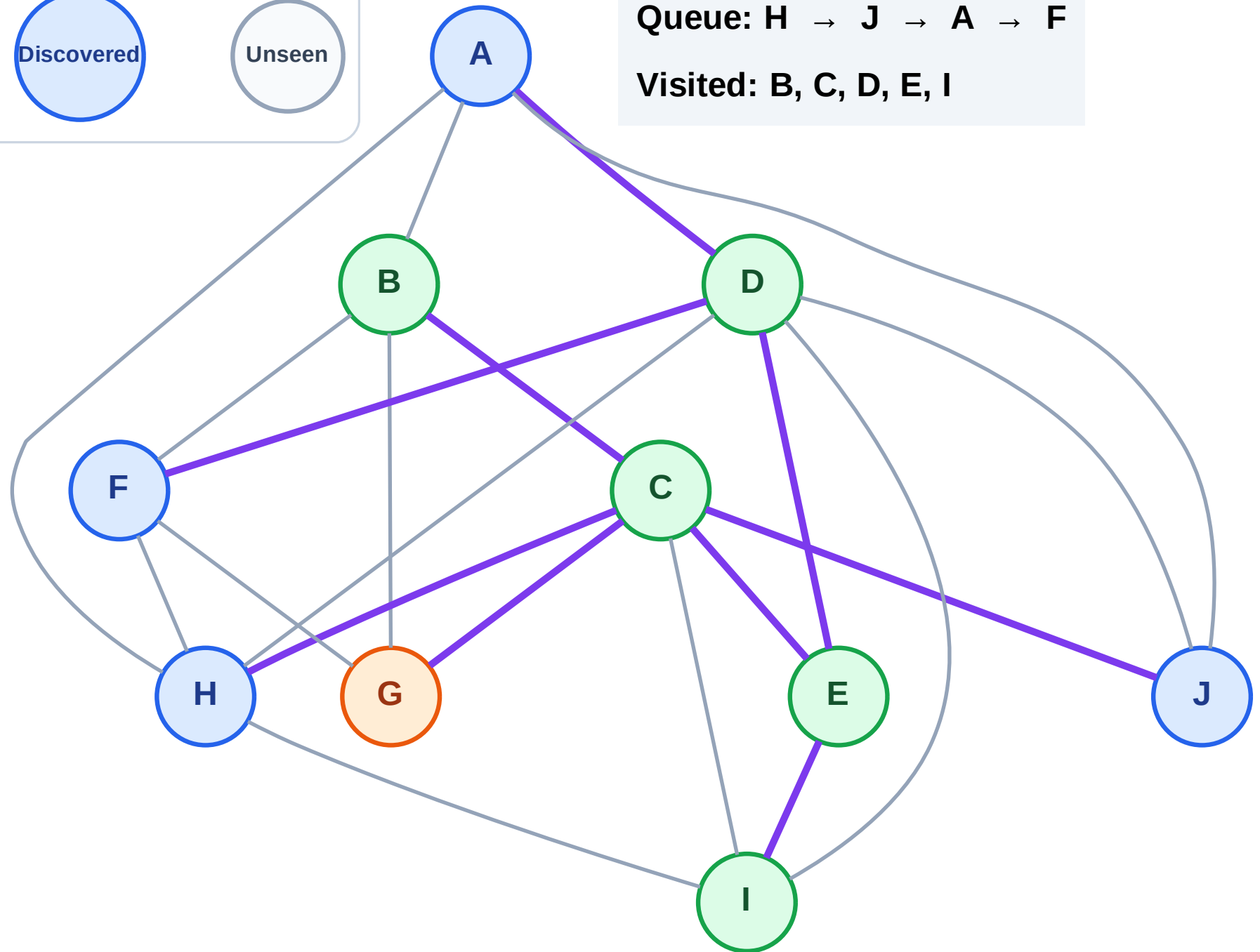
Processing

Discovered

Unseen

Queue: H → J → A → F

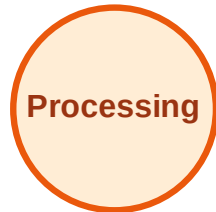
Visited: B, C, D, E, I



# BFS Traversal

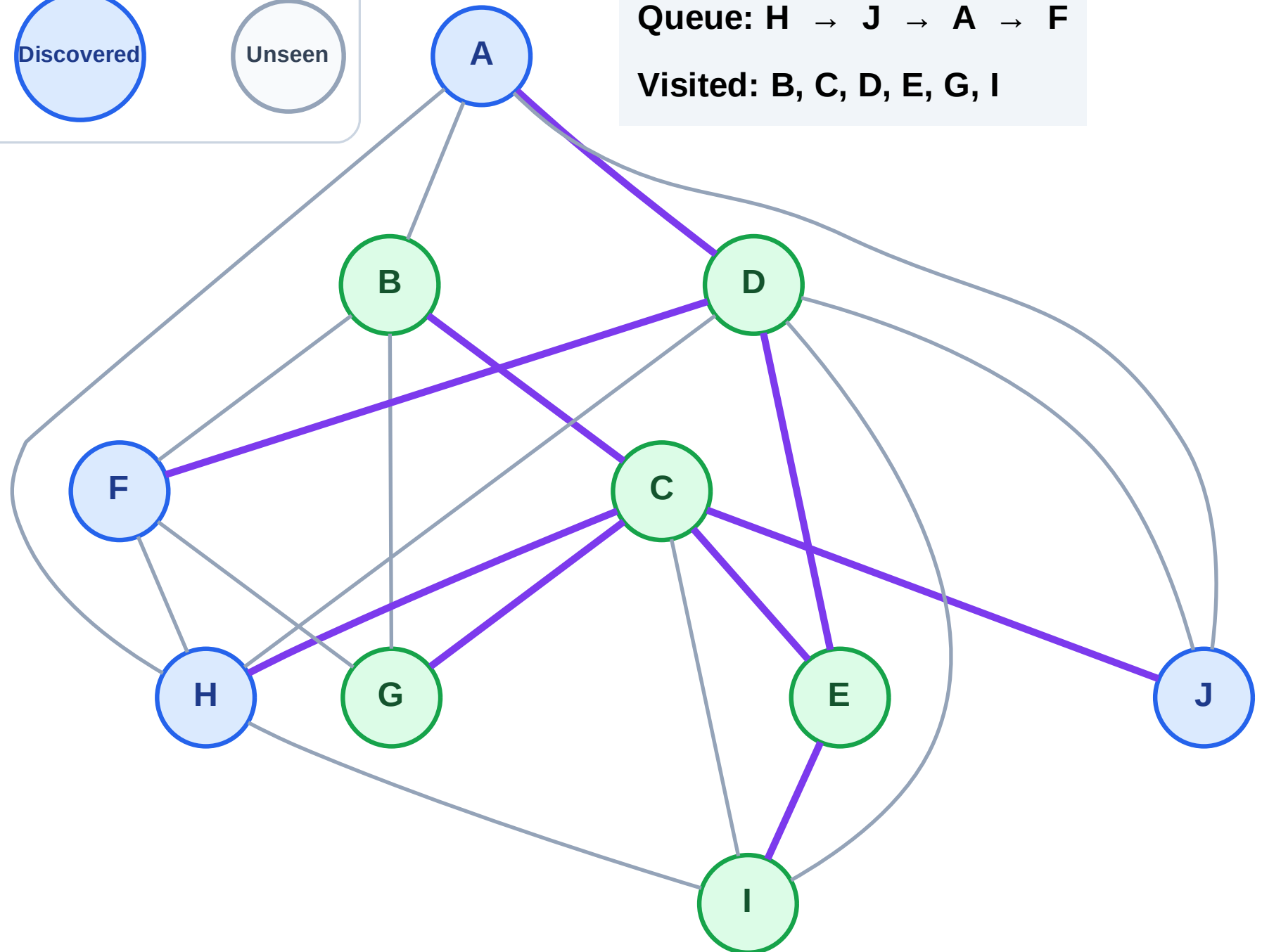
Step 13: No new neighbors from G

## Legend



Queue: H → J → A → F

Visited: B, C, D, E, G, I



# BFS Traversal

Step 14: Process node H

## Legend

Complete

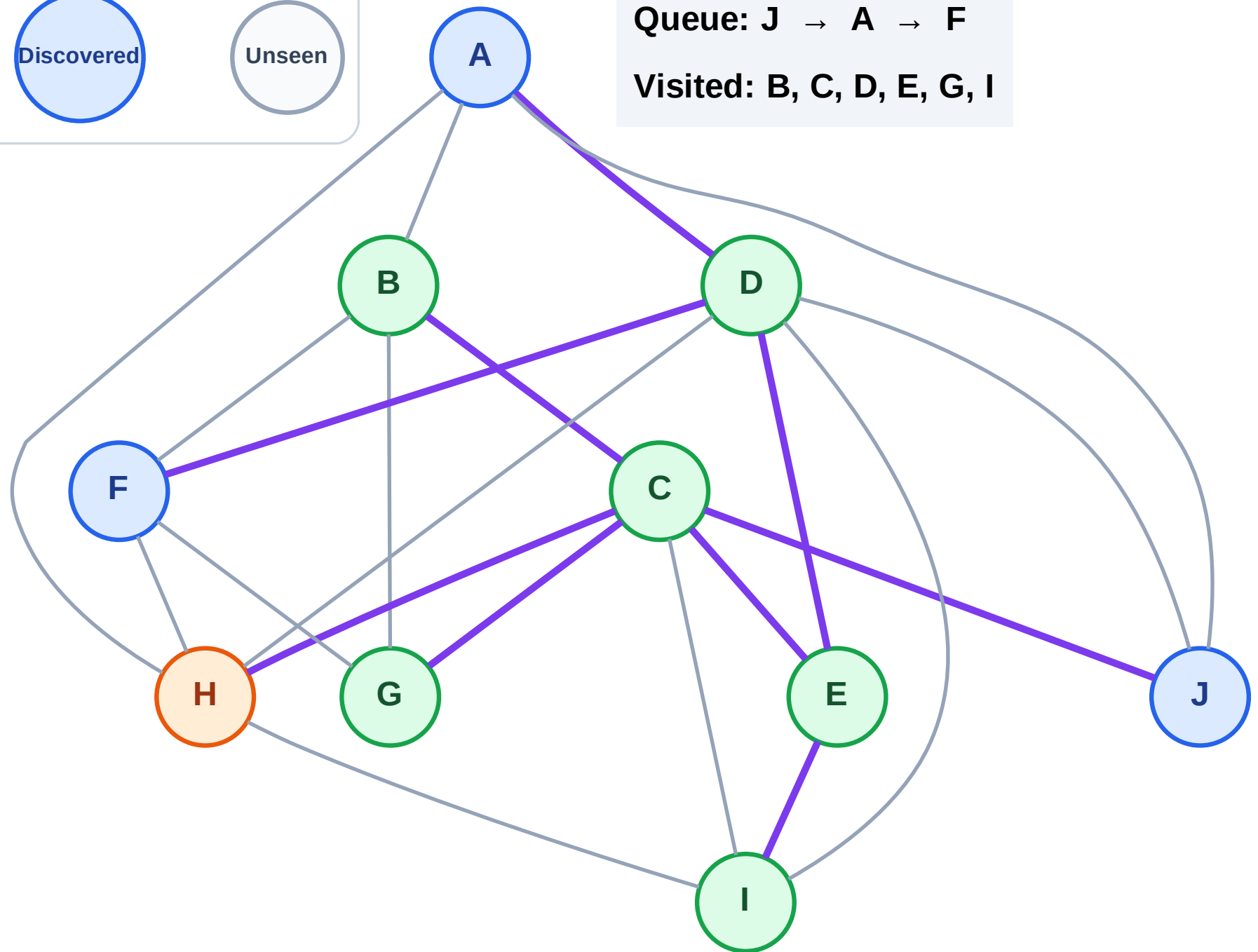
Processing

Discovered

Unseen

Queue: J → A → F

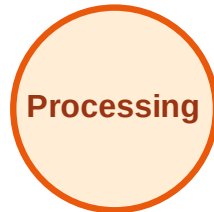
Visited: B, C, D, E, G, I



# BFS Traversal

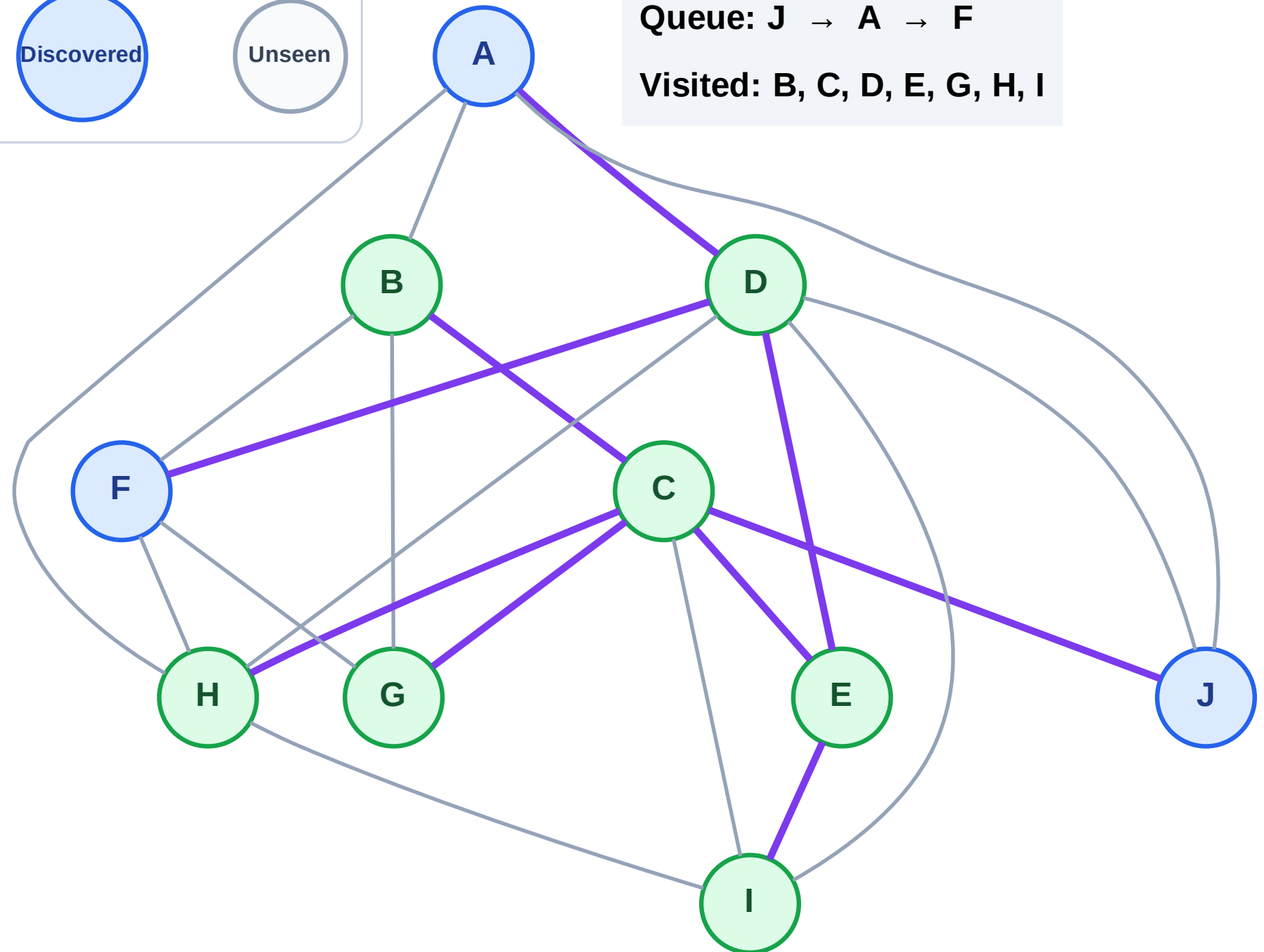
Step 15: No new neighbors from H

## Legend



Queue: J → A → F

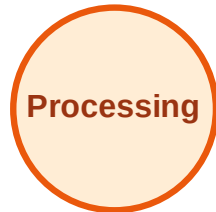
Visited: B, C, D, E, G, H, I



# BFS Traversal

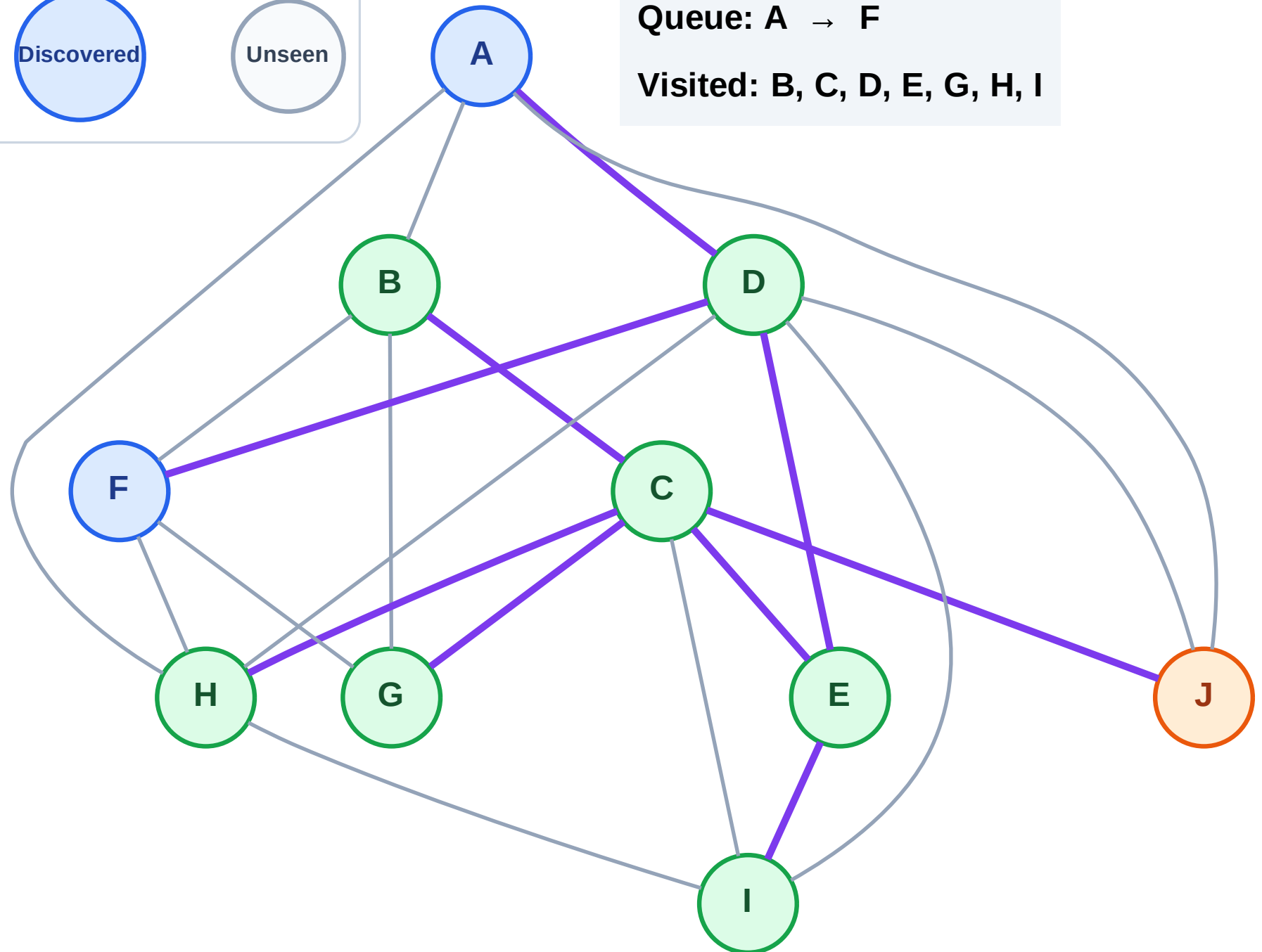
Step 16: Process node J

## Legend



Queue: A → F

Visited: B, C, D, E, G, H, I

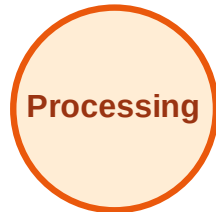




# BFS Traversal

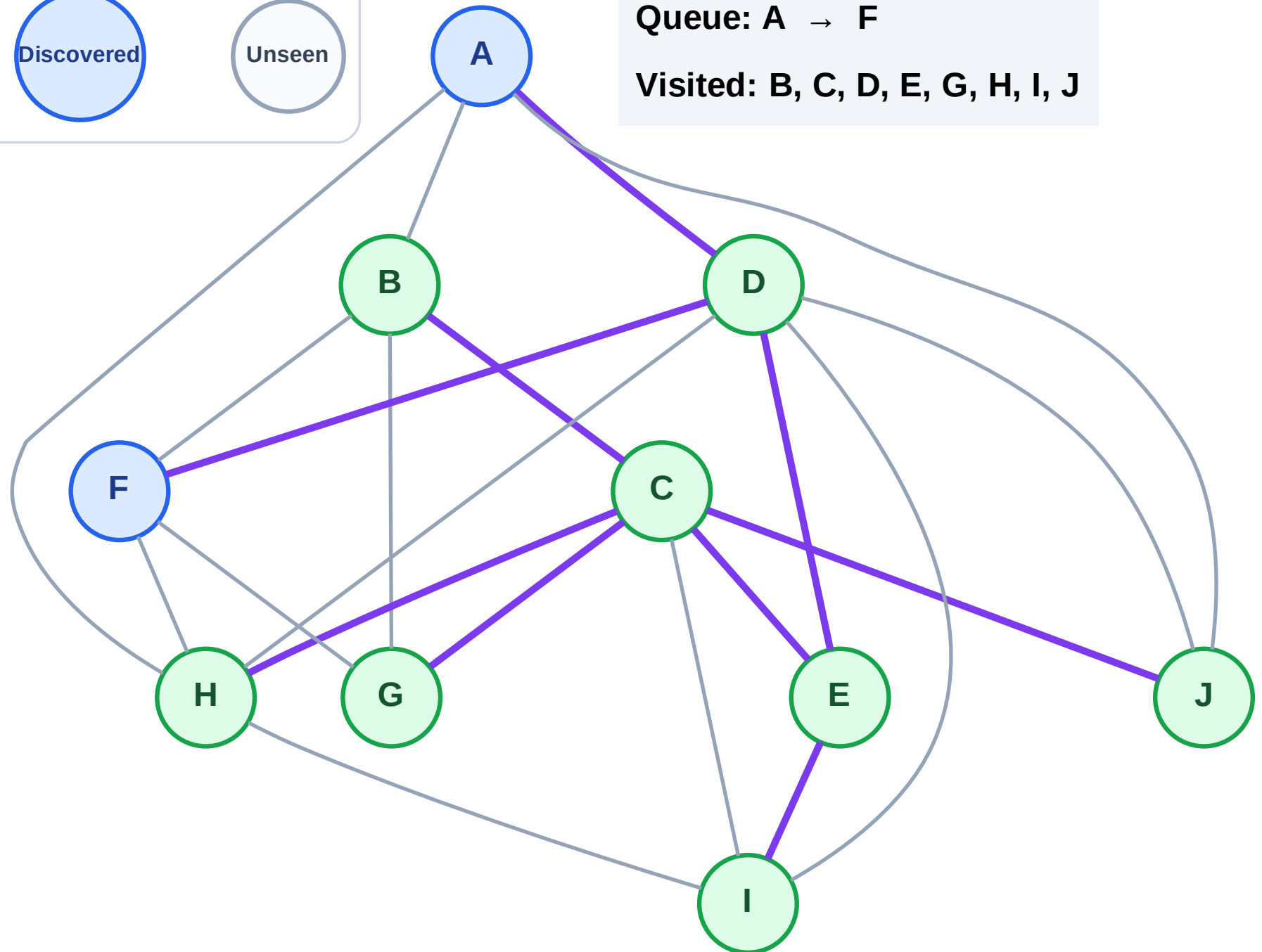
Step 17: No new neighbors from J

## Legend



Queue: A → F

Visited: B, C, D, E, G, H, I, J



# BFS Traversal

Step 18: Process node A

## Legend

Complete

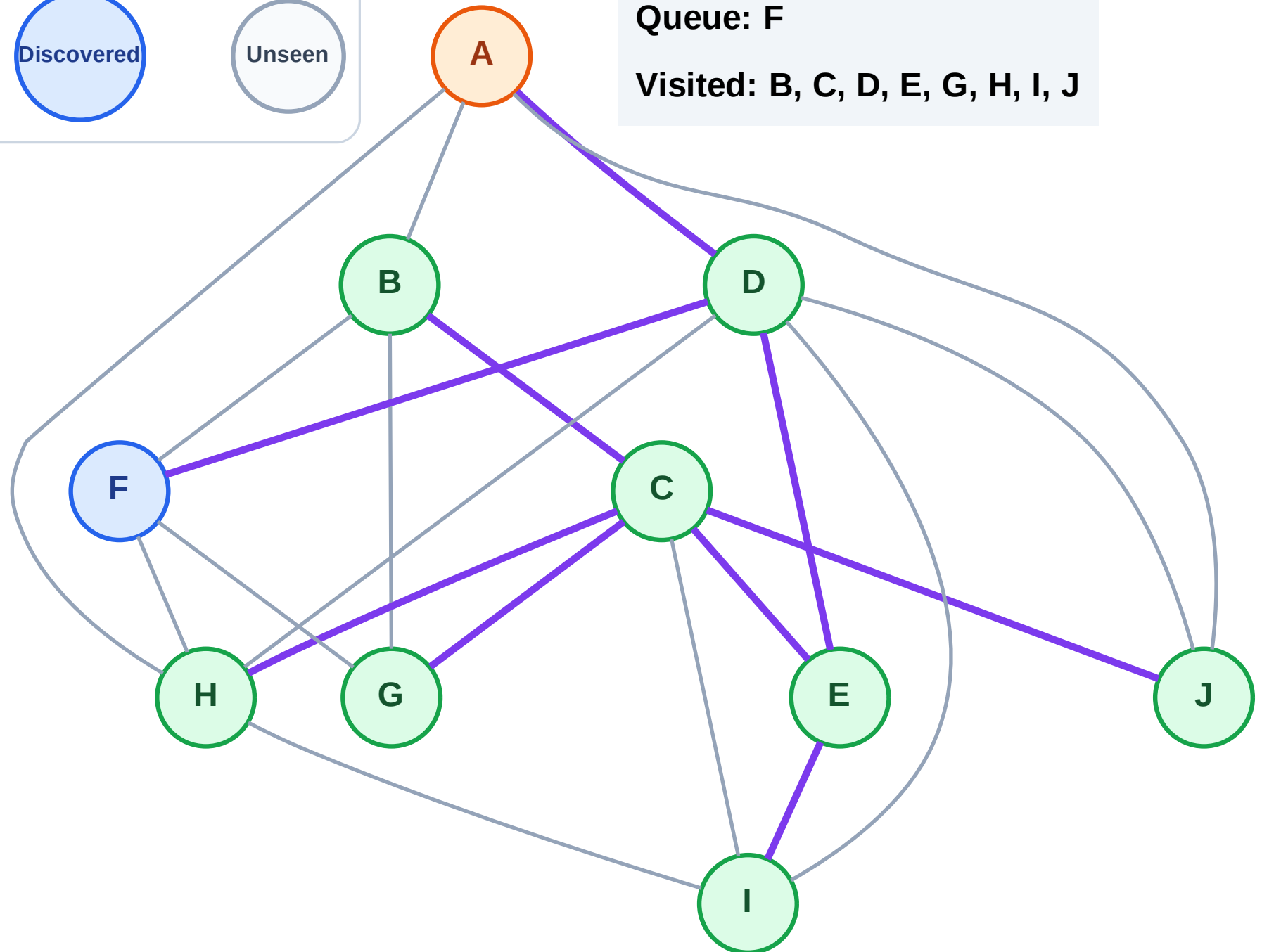
Processing

Discovered

Unseen

Queue: F

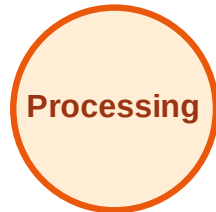
Visited: B, C, D, E, G, H, I, J



# BFS Traversal

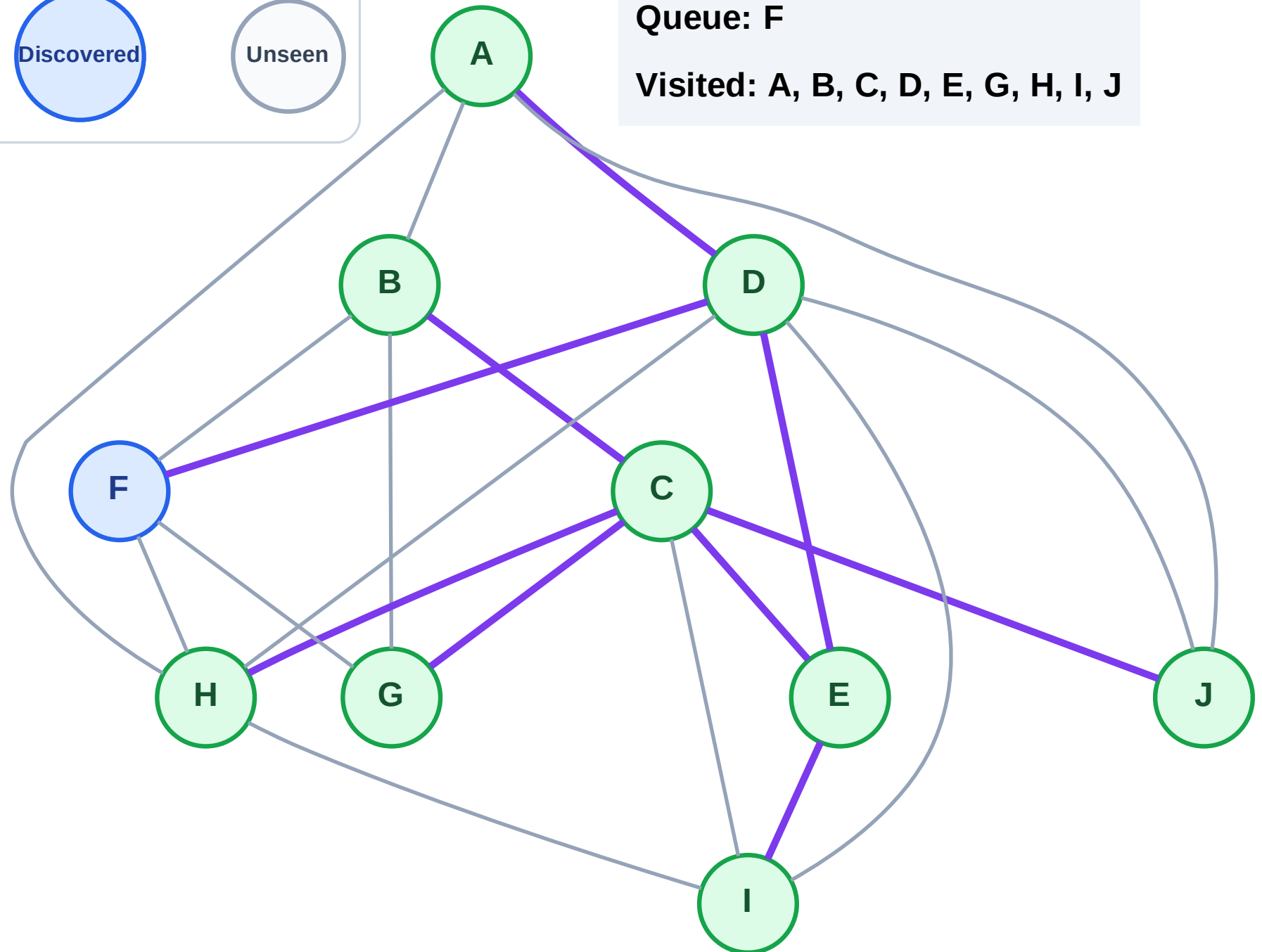
Step 19: No new neighbors from A

## Legend



Queue: F

Visited: A, B, C, D, E, G, H, I, J



# BFS Traversal

Step 20: Process node F

## Legend

Complete

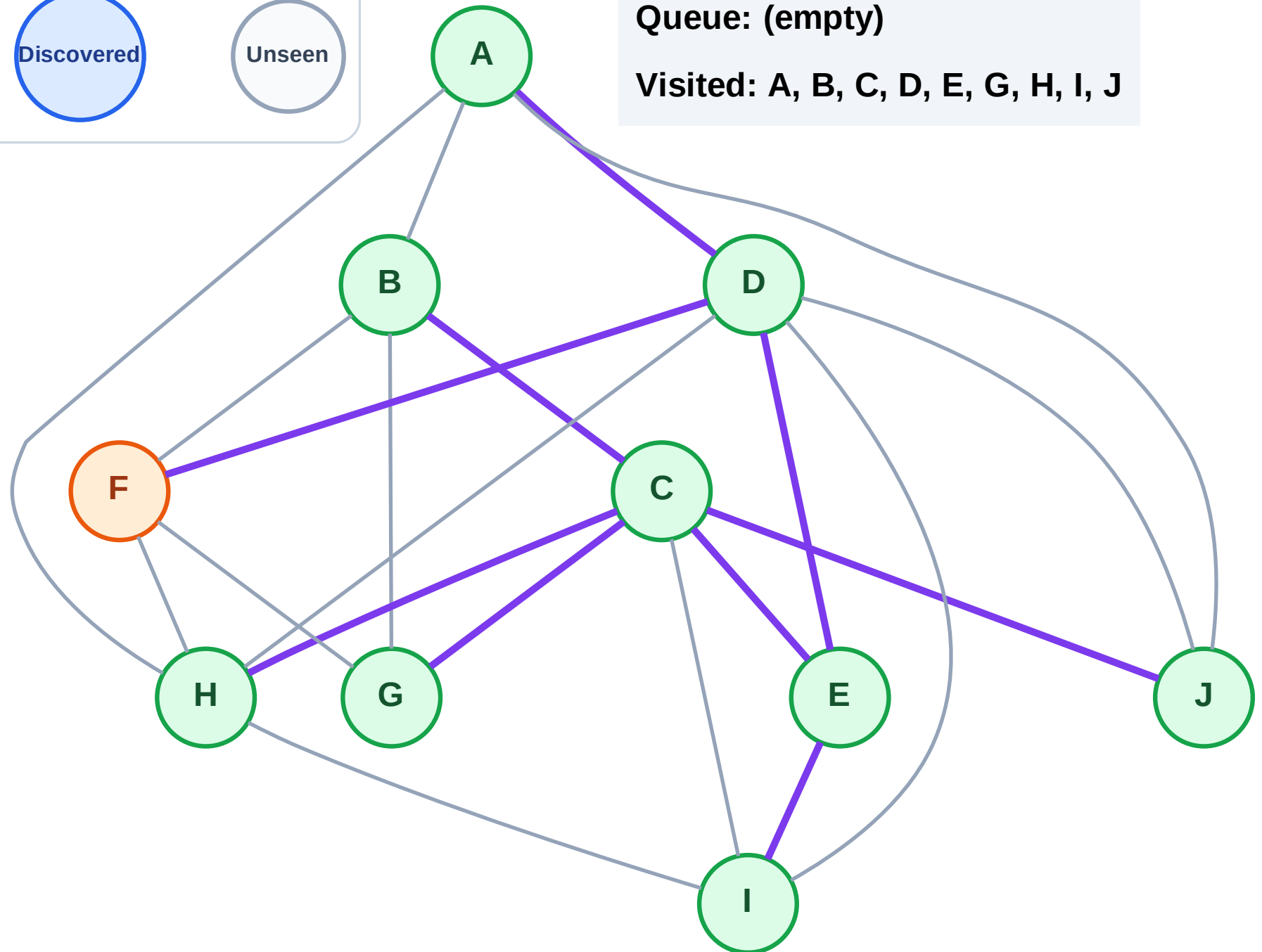
Processing

Discovered

Unseen

Queue: (empty)

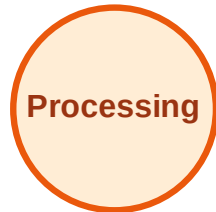
Visited: A, B, C, D, E, G, H, I, J



# BFS Traversal

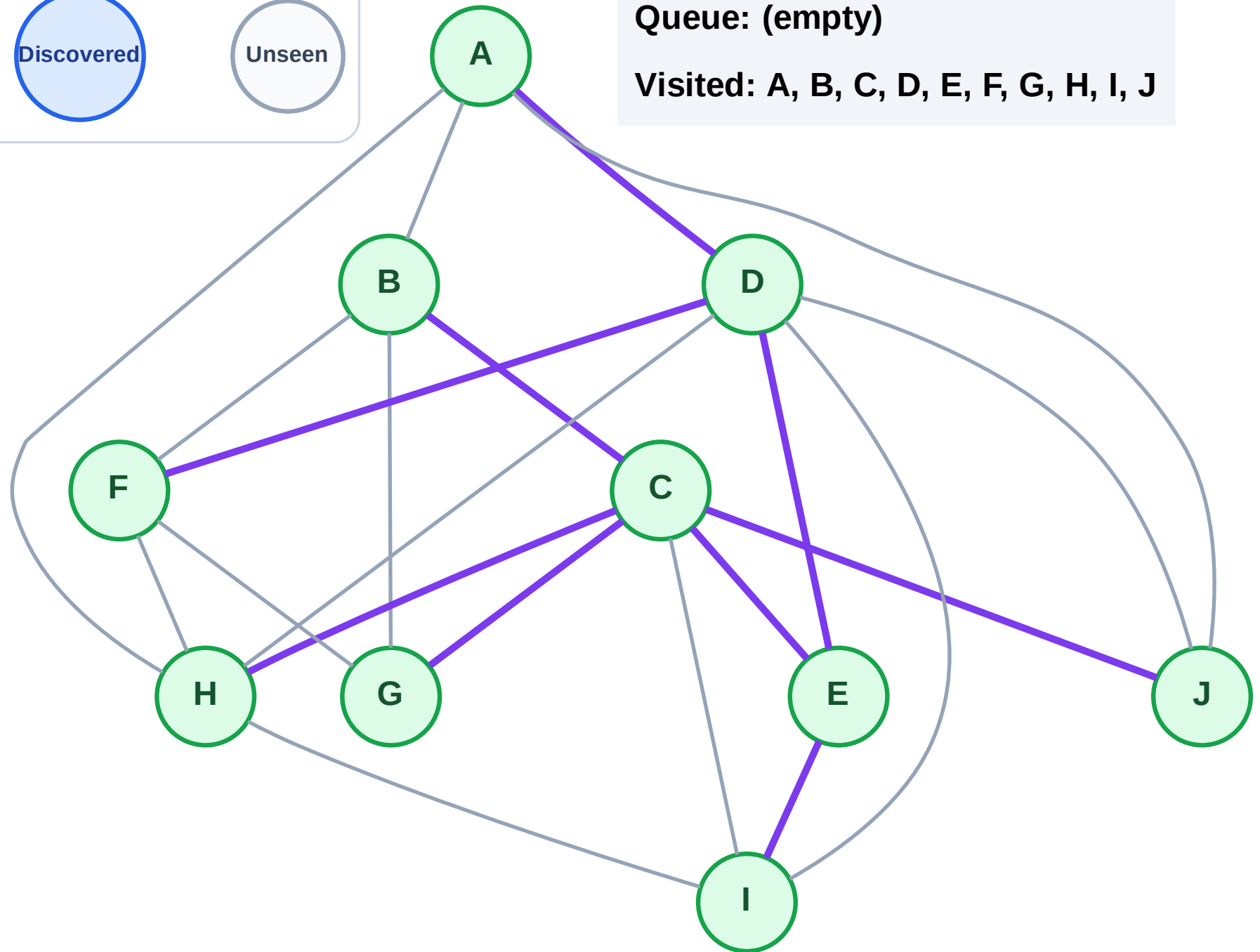
Step 21: No new neighbors from F

## Legend



Queue: (empty)

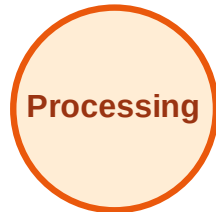
Visited: A, B, C, D, E, F, G, H, I, J



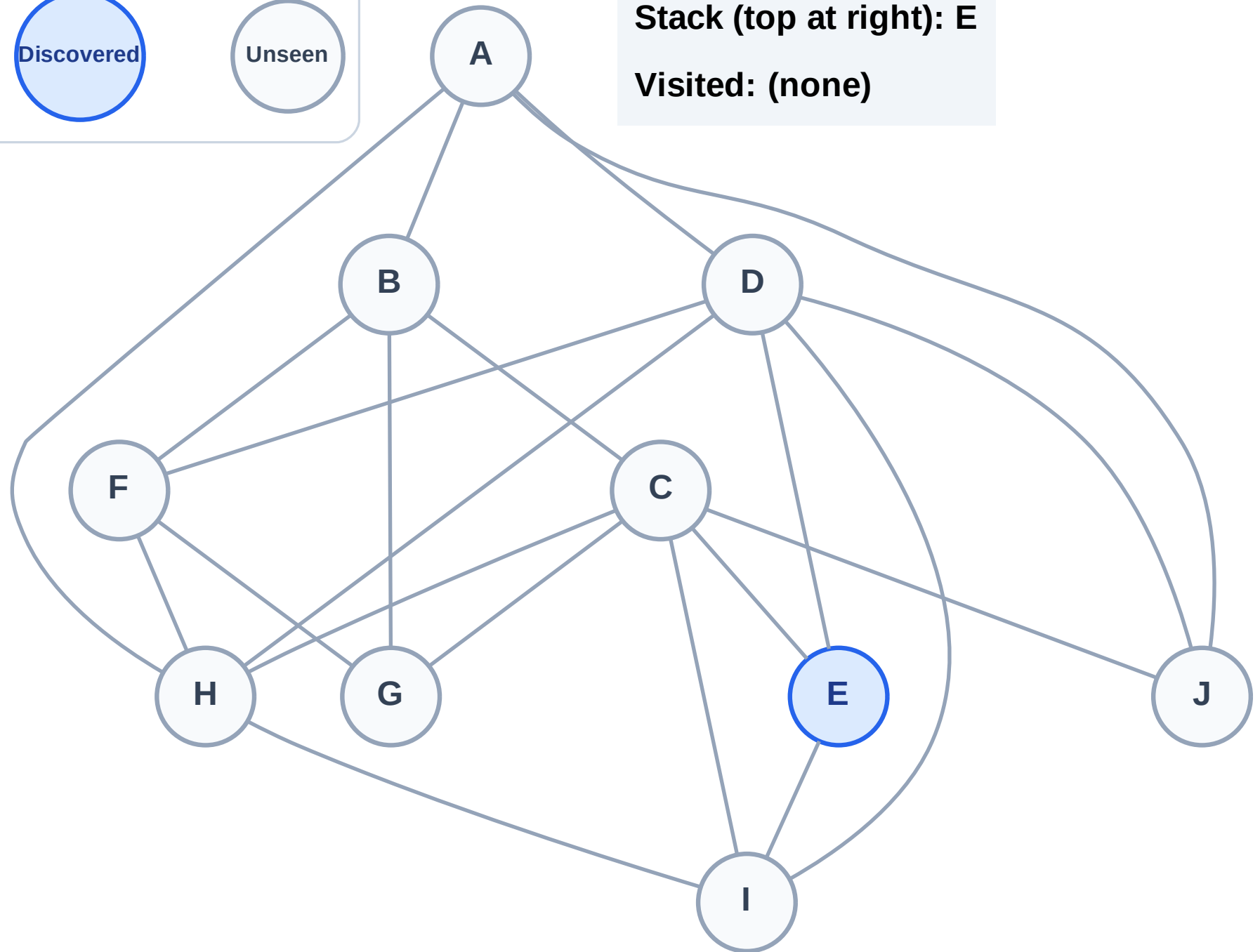
# DFS Traversal

Step 1: Start at E

## Legend



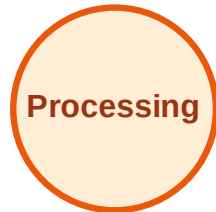
Stack (top at right): E  
Visited: (none)



# DFS Traversal

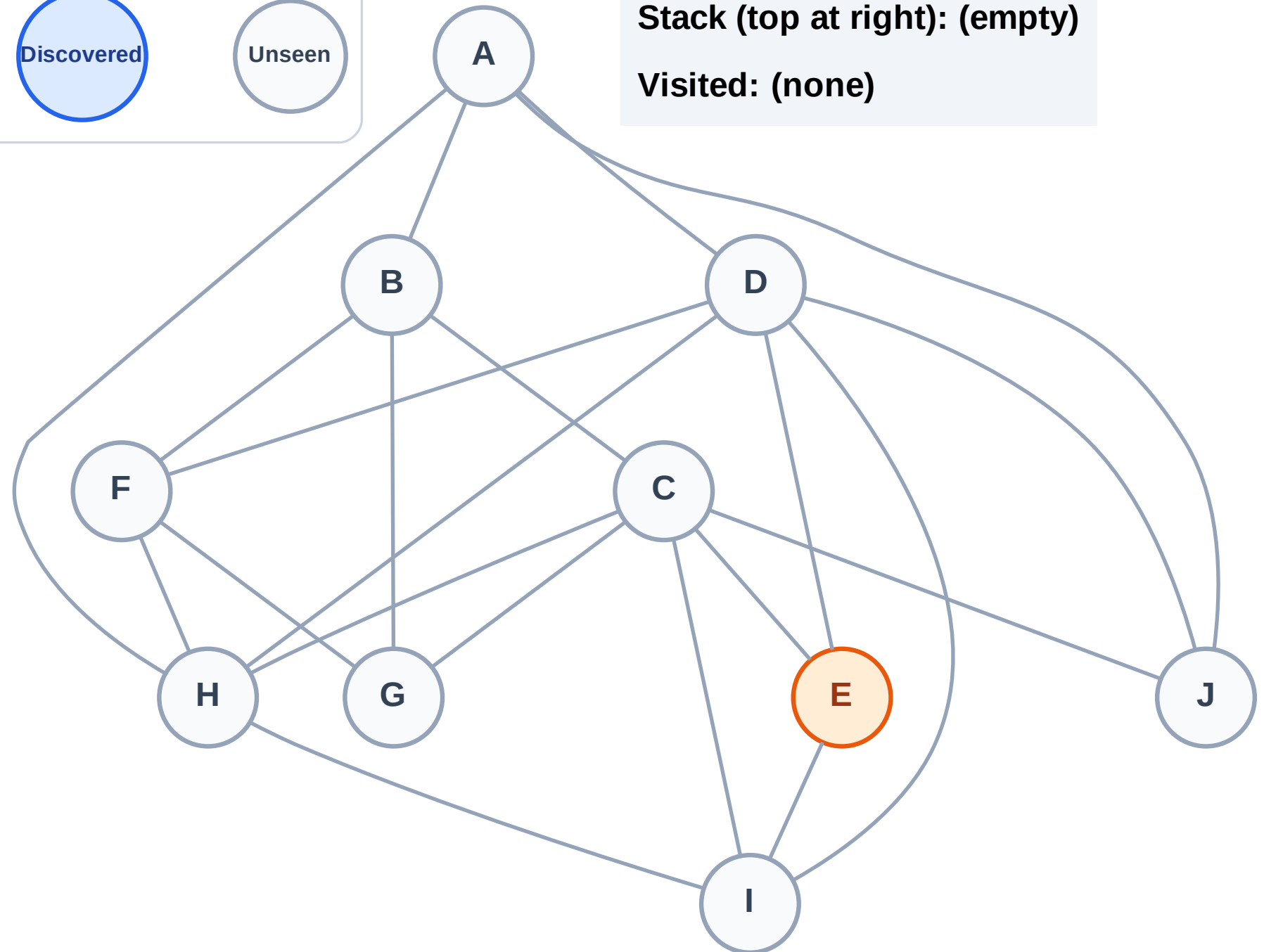
Step 2: Process node E

## Legend



Stack (top at right): (empty)

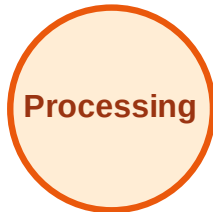
Visited: (none)



# DFS Traversal

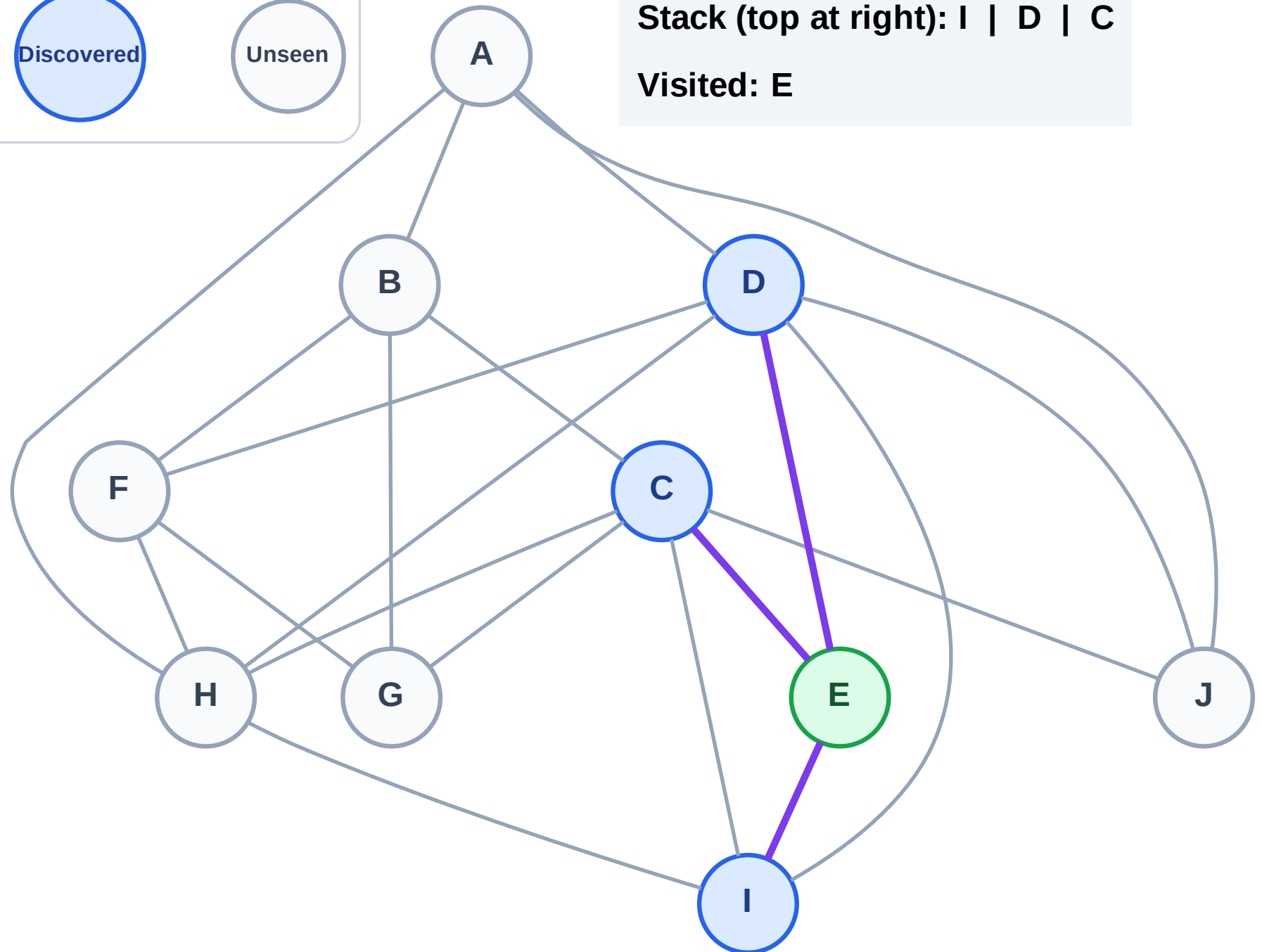
Step 3: Discover I, D, C from E

## Legend



Stack (top at right): I | D | C

Visited: E

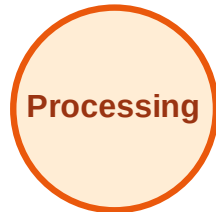




# DFS Traversal

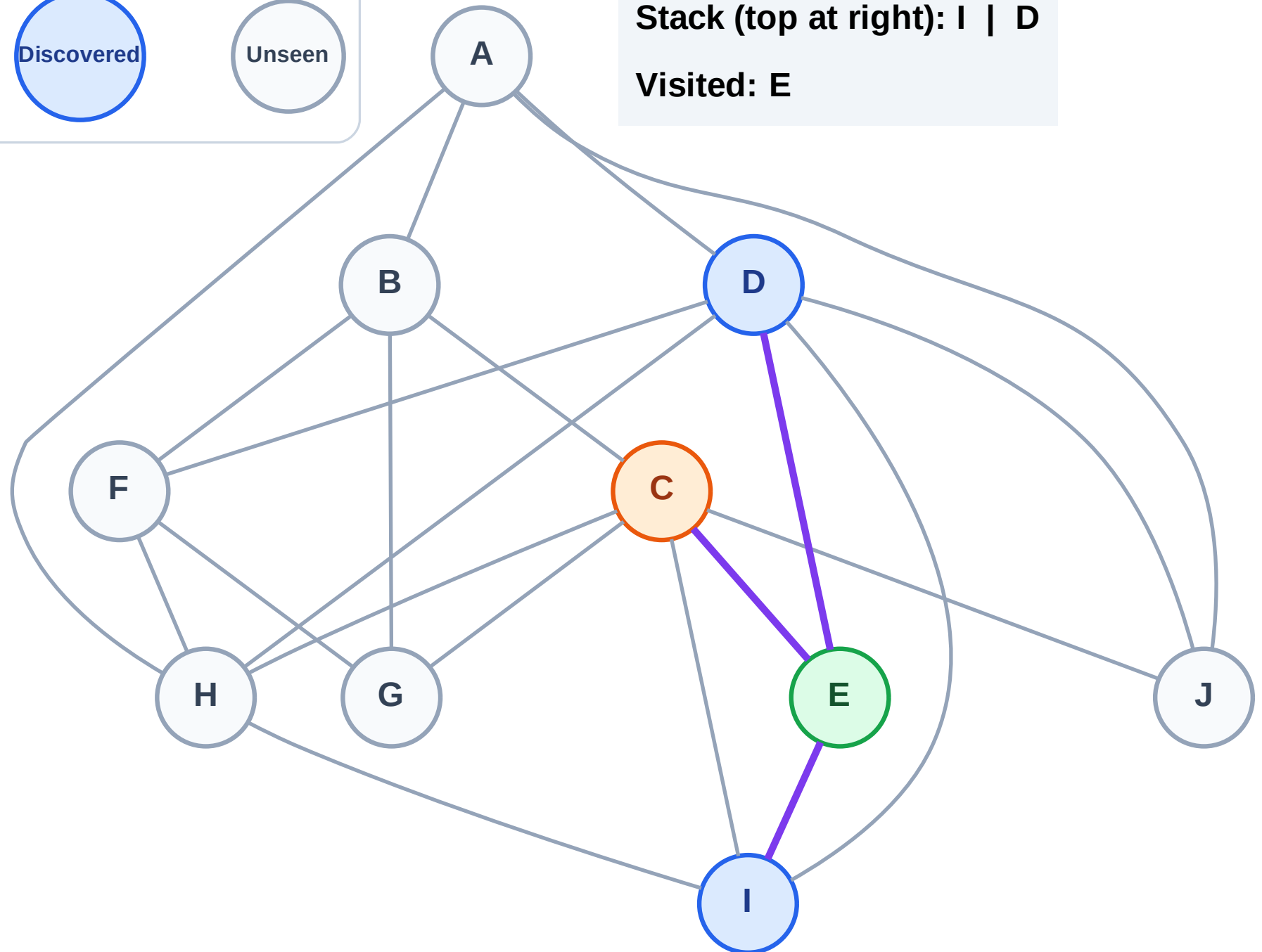
Step 4: Process node C

## Legend



Stack (top at right): I | D

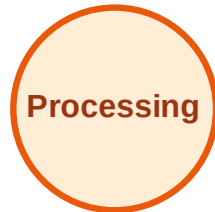
Visited: E



# DFS Traversal

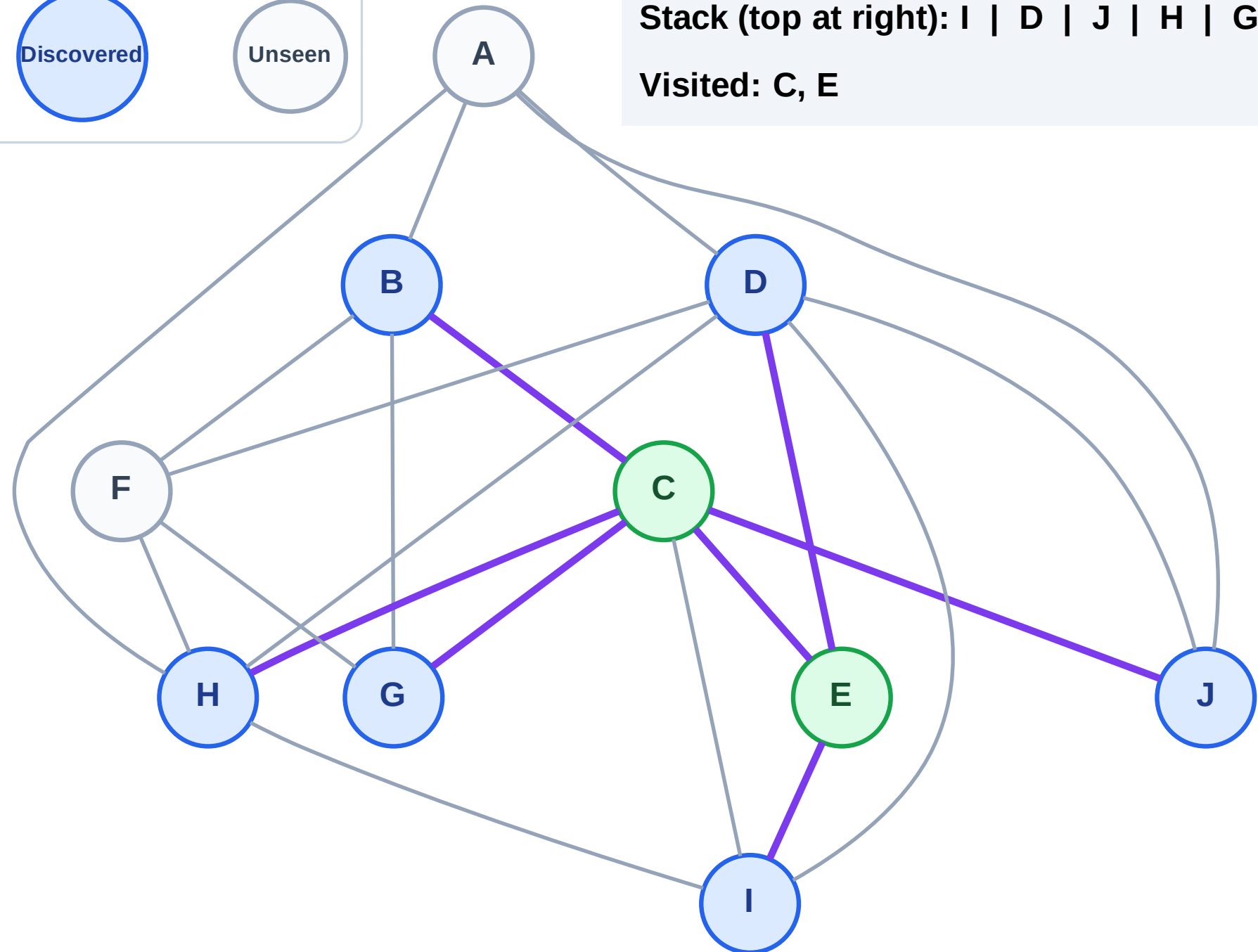
Step 5: Discover J, H, G, B from C

## Legend



Stack (top at right): I | D | J | H | G | B

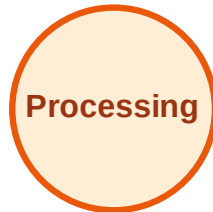
Visited: C, E



# DFS Traversal

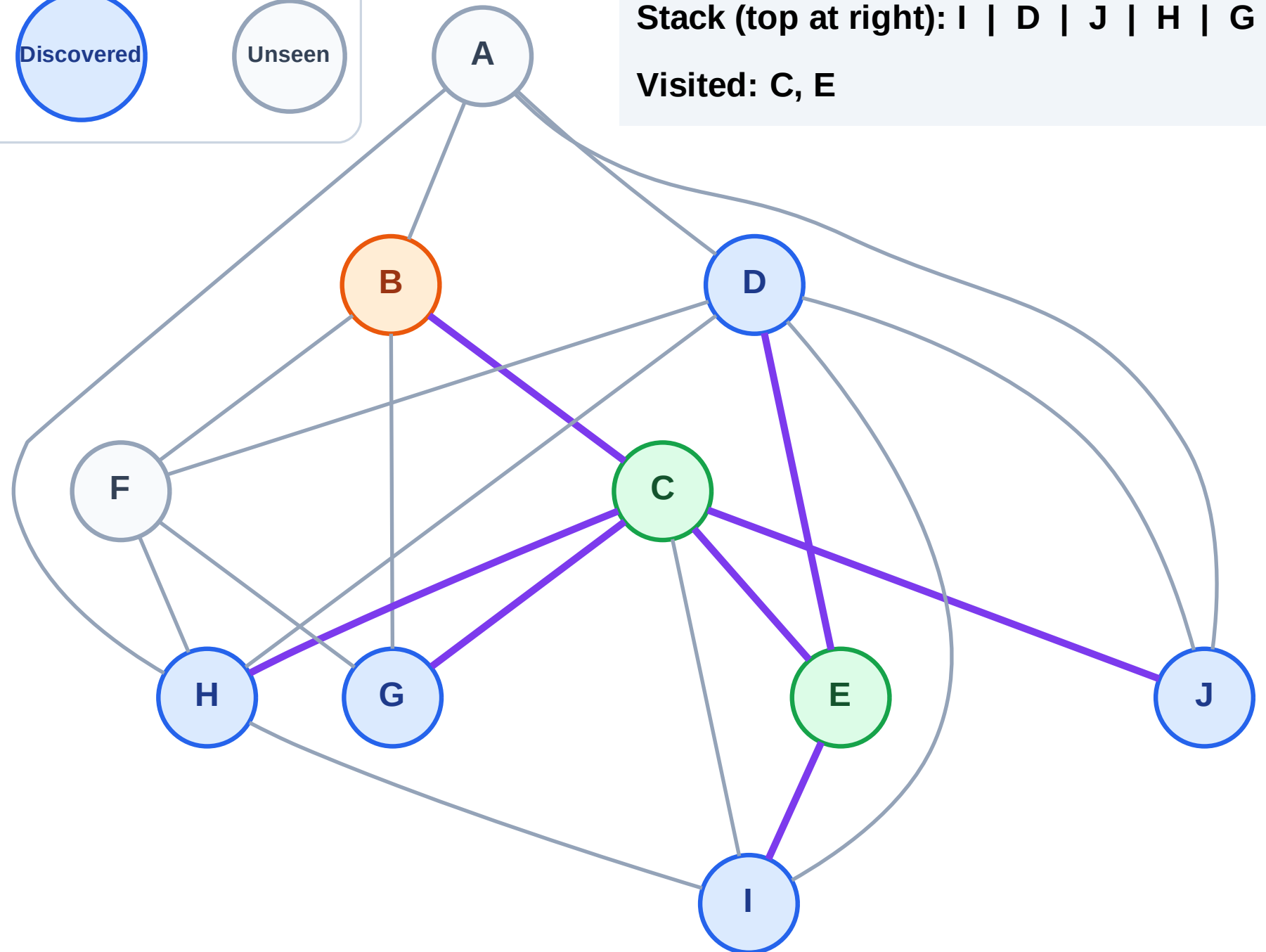
Step 6: Process node B

## Legend



Stack (top at right): I | D | J | H | G

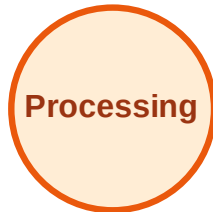
Visited: C, E



# DFS Traversal

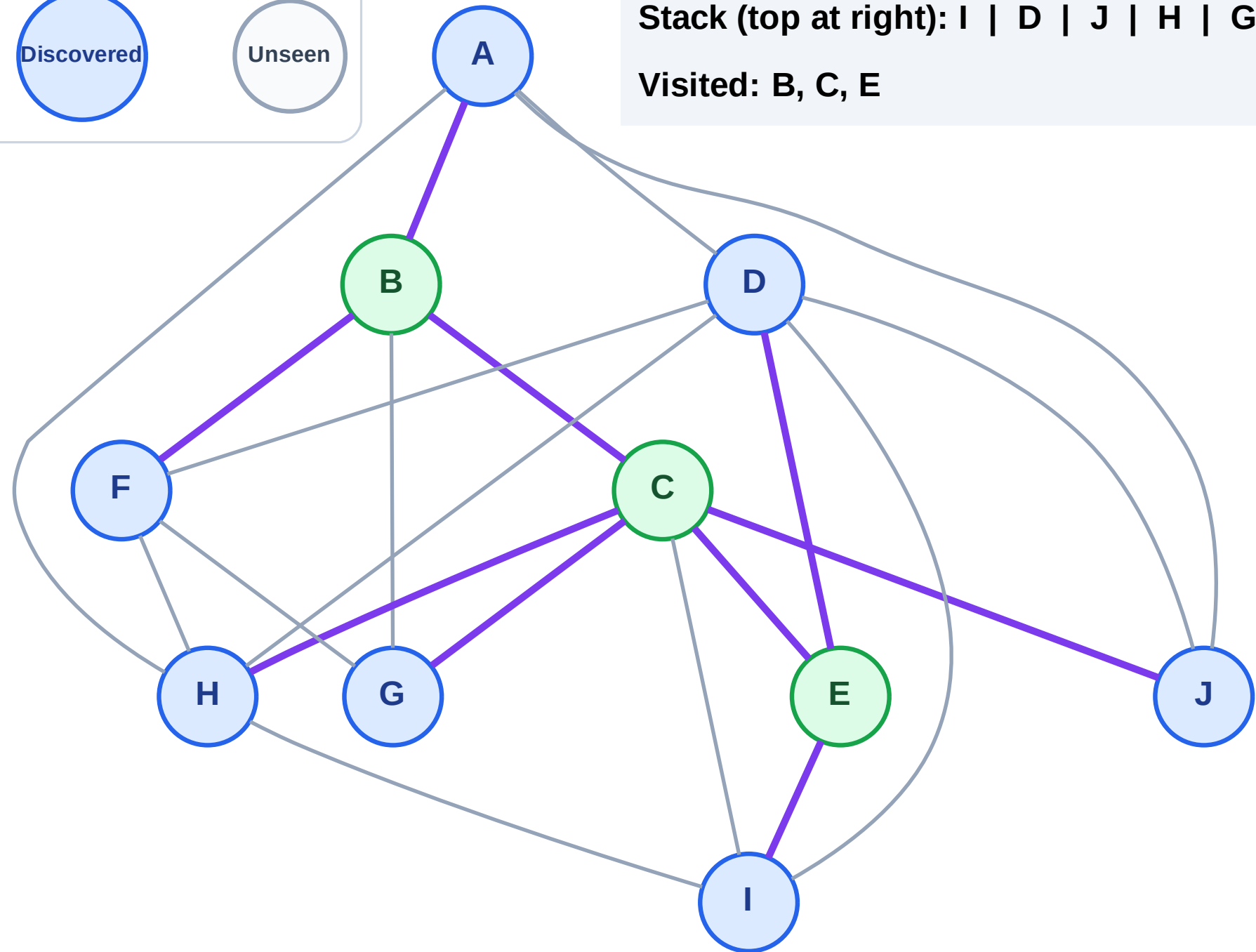
Step 7: Discover F, A from B

## Legend



Stack (top at right): I | D | J | H | G | F | A

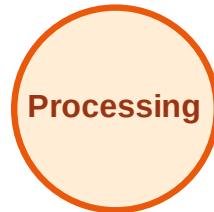
Visited: B, C, E



# DFS Traversal

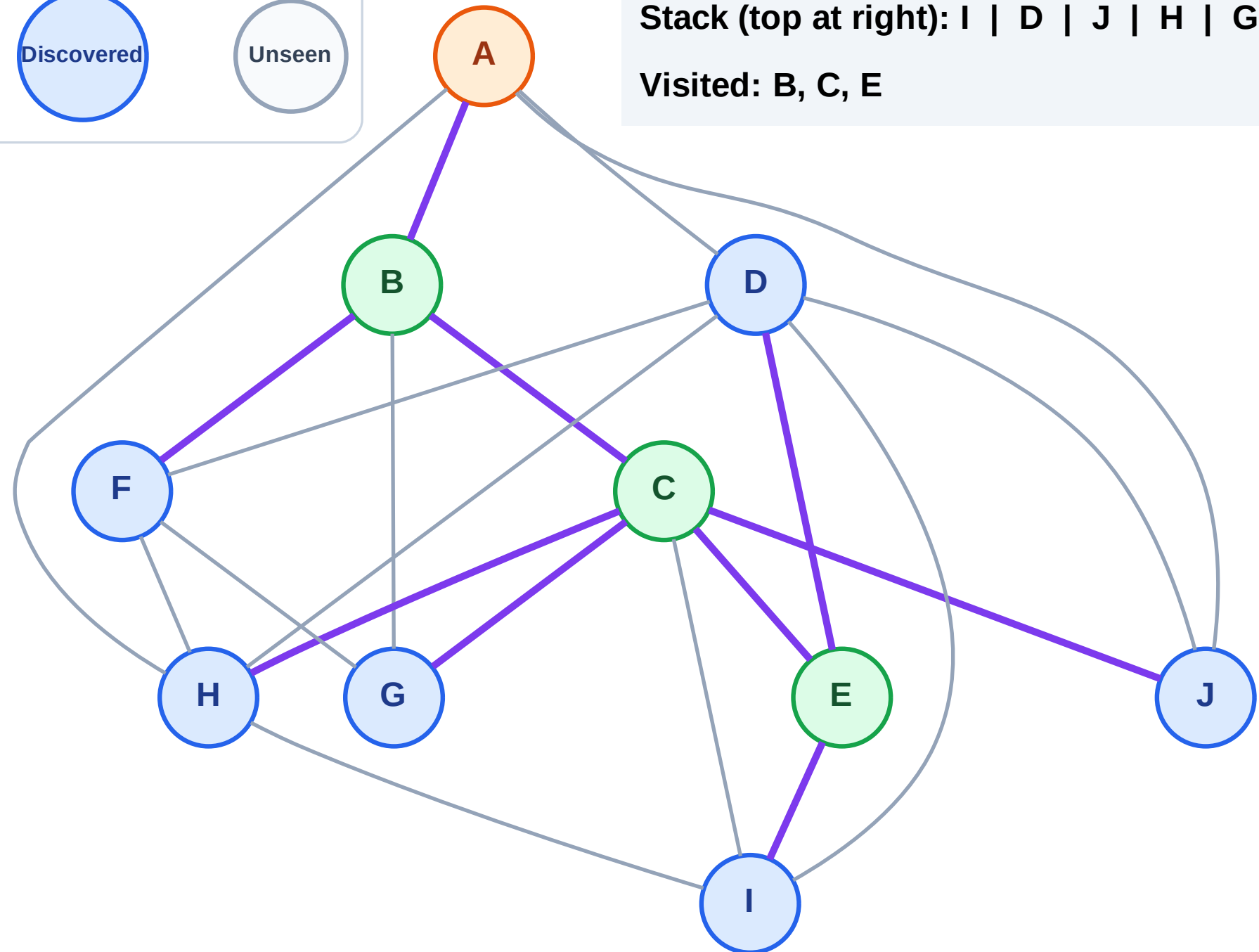
Step 8: Process node A

## Legend



Stack (top at right): I | D | J | H | G | F

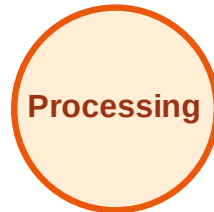
Visited: B, C, E



# DFS Traversal

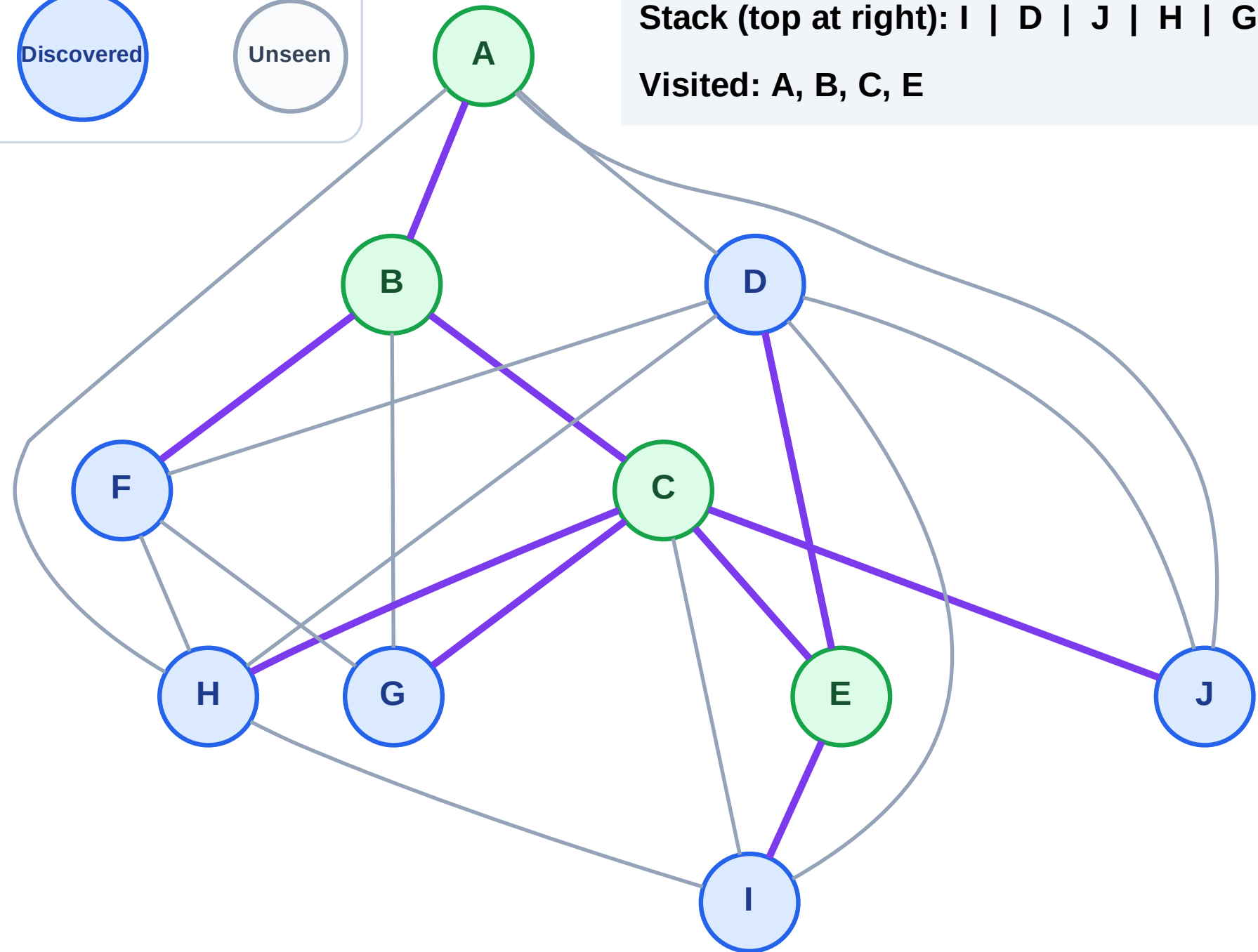
Step 9: No new neighbors from A

## Legend



Stack (top at right): I | D | J | H | G | F

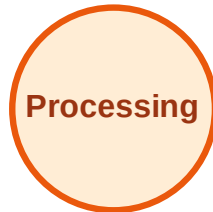
Visited: A, B, C, E



# DFS Traversal

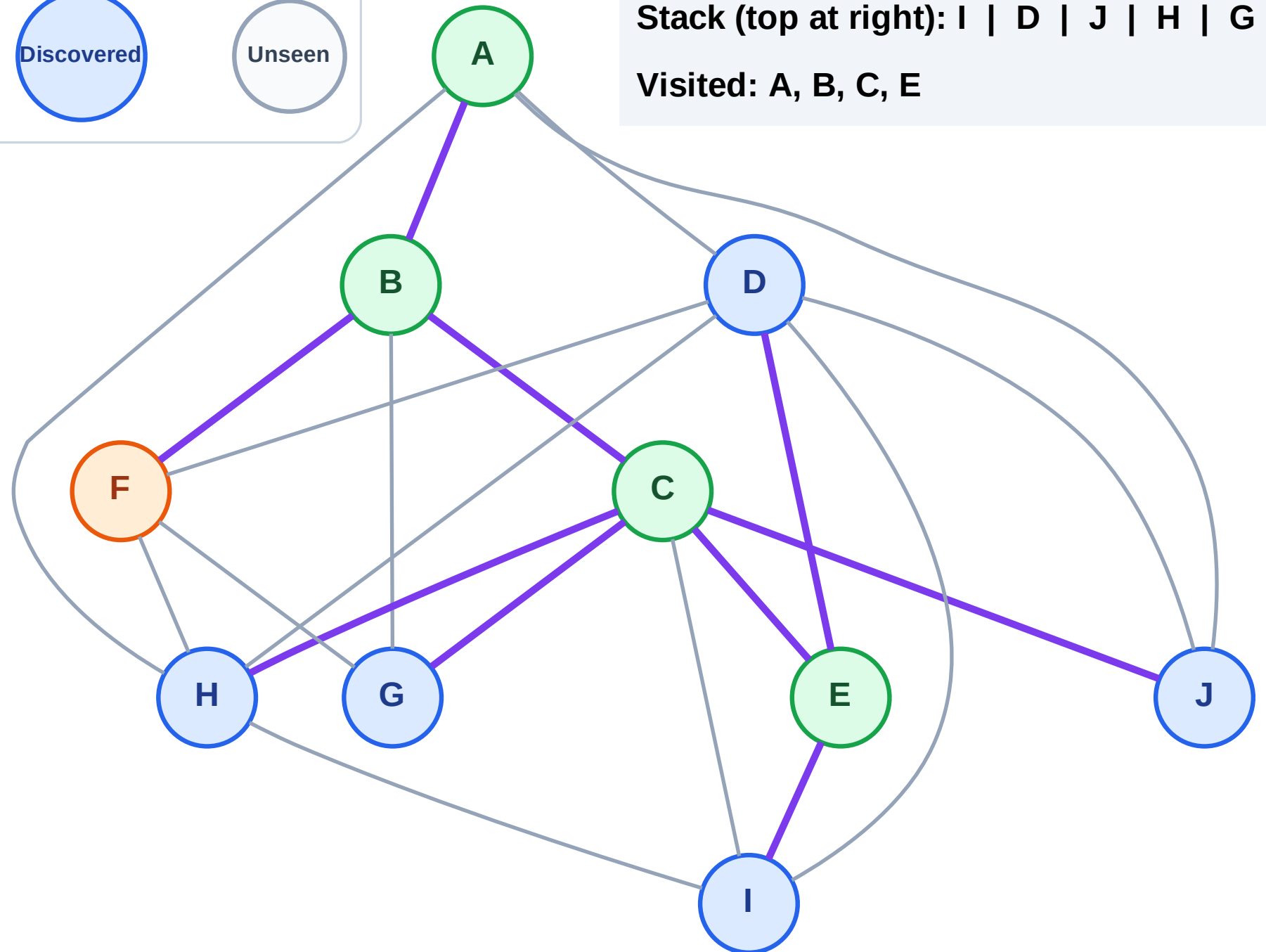
Step 10: Process node F

## Legend



Stack (top at right): I | D | J | H | G

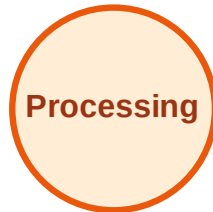
Visited: A, B, C, E



# DFS Traversal

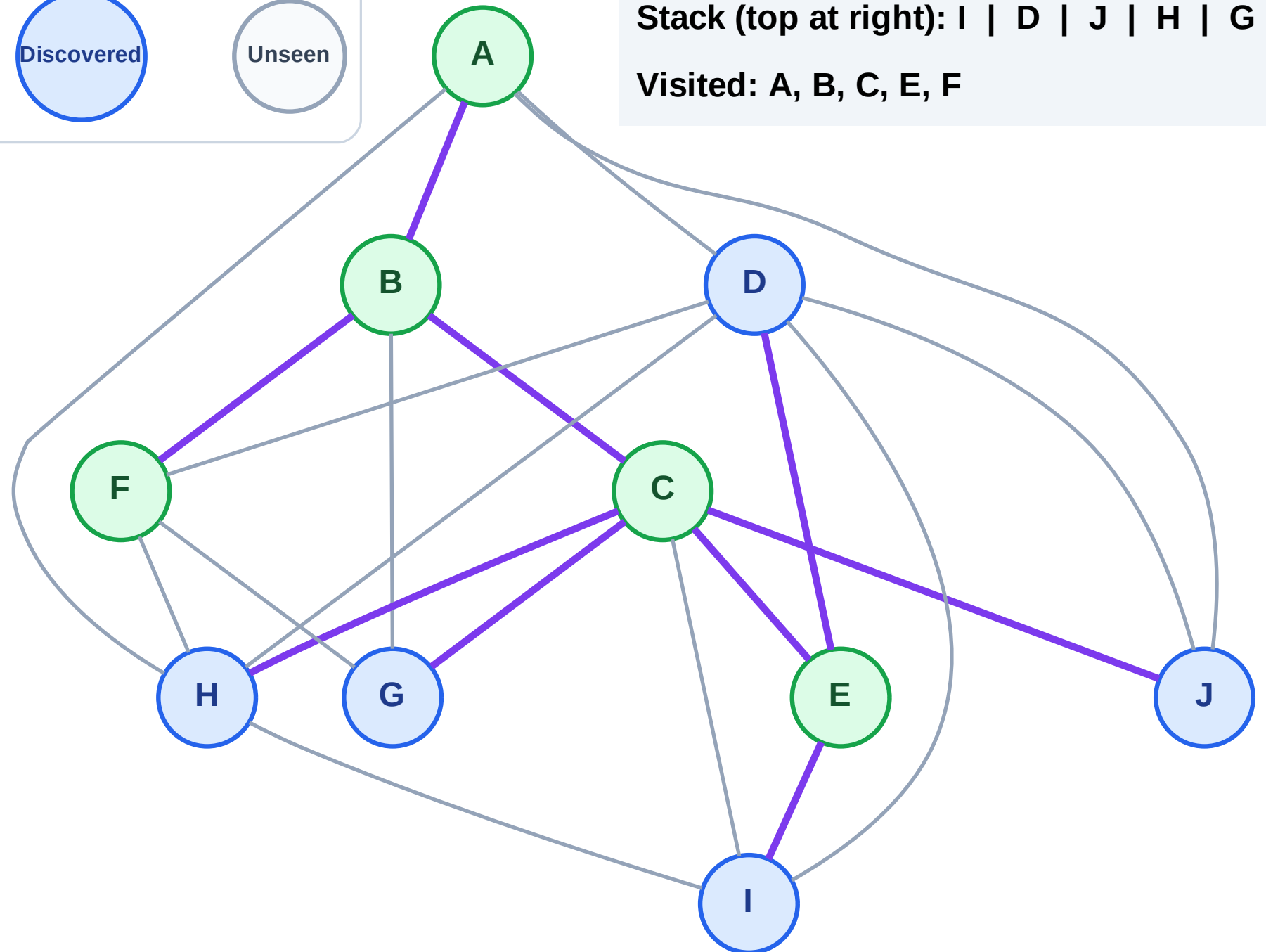
Step 11: No new neighbors from F

## Legend



Stack (top at right): I | D | J | H | G

Visited: A, B, C, E, F





# DFS Traversal

Step 12: Process node G

## Legend

Complete

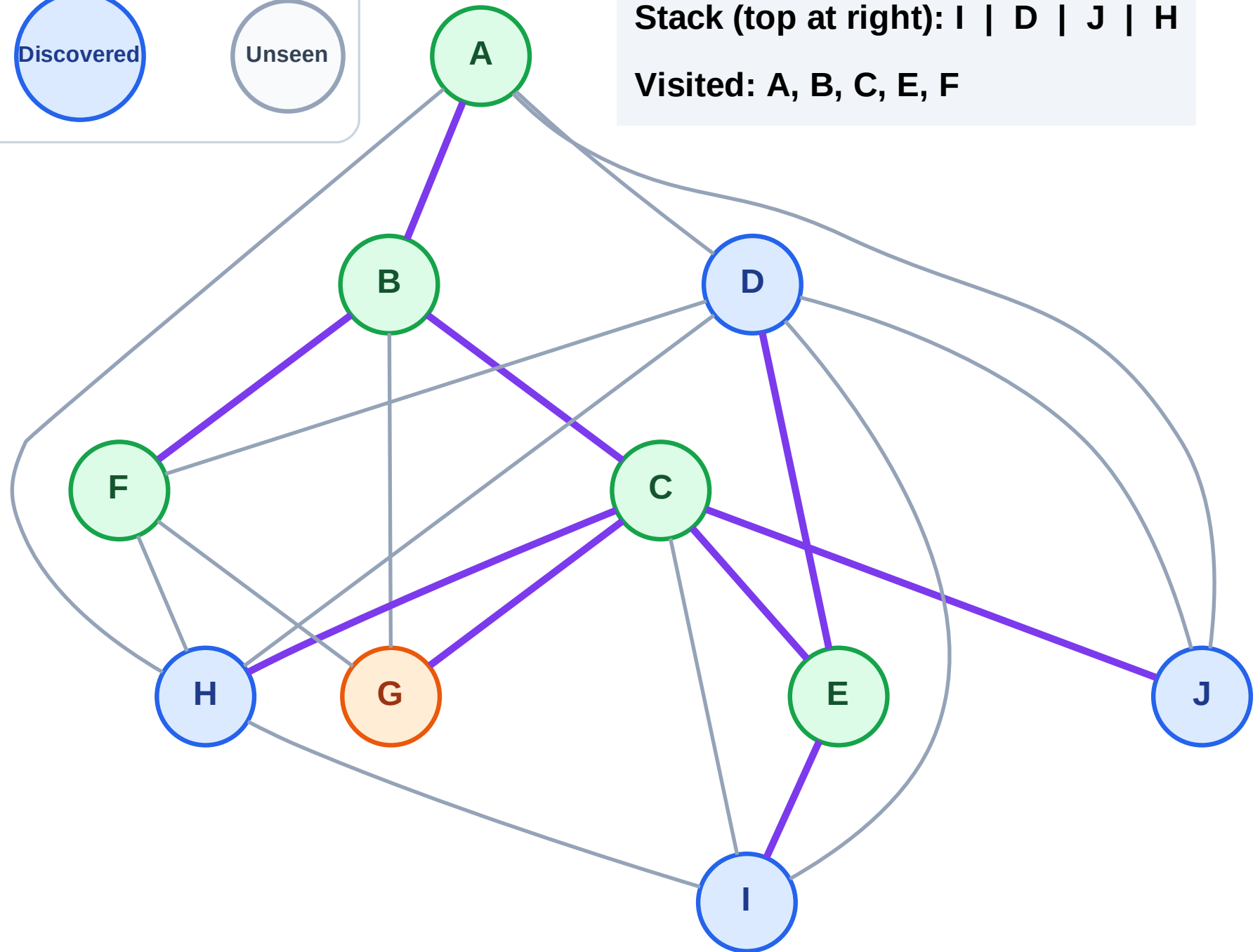
Processing

Discovered

Unseen

Stack (top at right): I | D | J | H

Visited: A, B, C, E, F



# DFS Traversal

Step 13: No new neighbors from G

## Legend

Complete

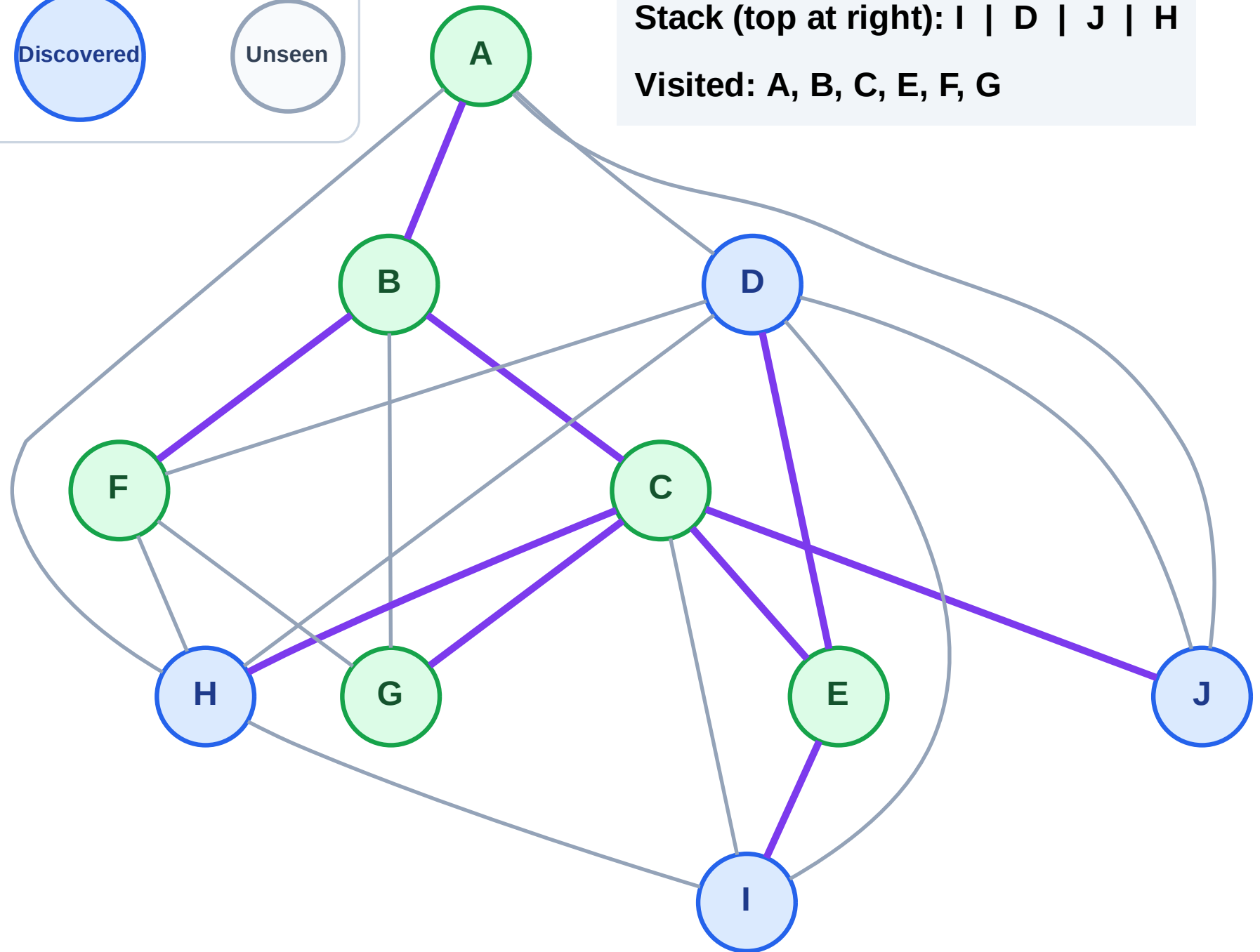
Processing

Discovered

Unseen

Stack (top at right): I | D | J | H

Visited: A, B, C, E, F, G



# DFS Traversal

Step 14: Process node H

## Legend

Complete

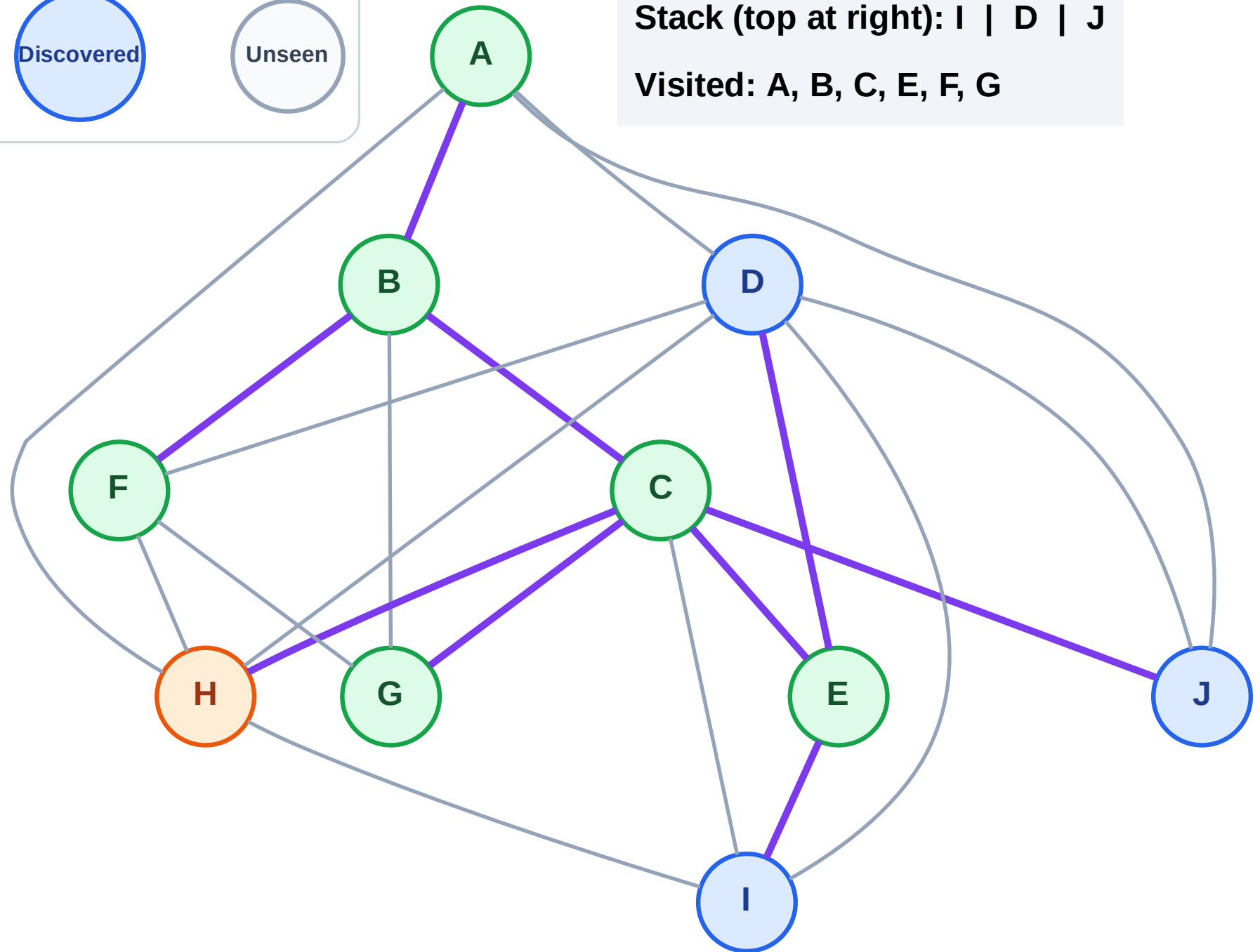
Processing

Discovered

Unseen

Stack (top at right): I | D | J

Visited: A, B, C, E, F, G



# DFS Traversal

Step 15: No new neighbors from H

## Legend

Complete

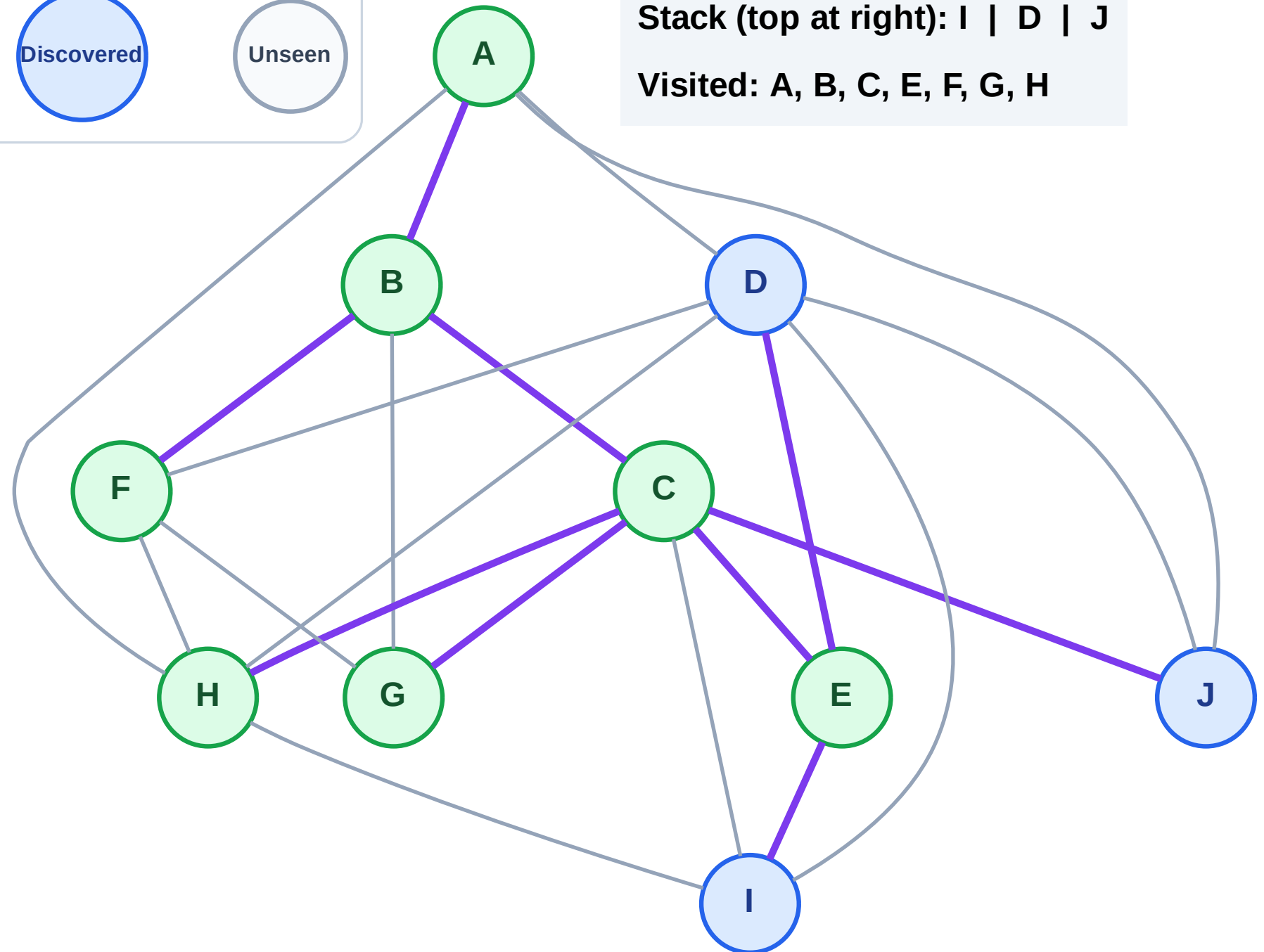
Processing

Discovered

Unseen

Stack (top at right): I | D | J

Visited: A, B, C, E, F, G, H



# DFS Traversal

Step 16: Process node J

## Legend

Complete

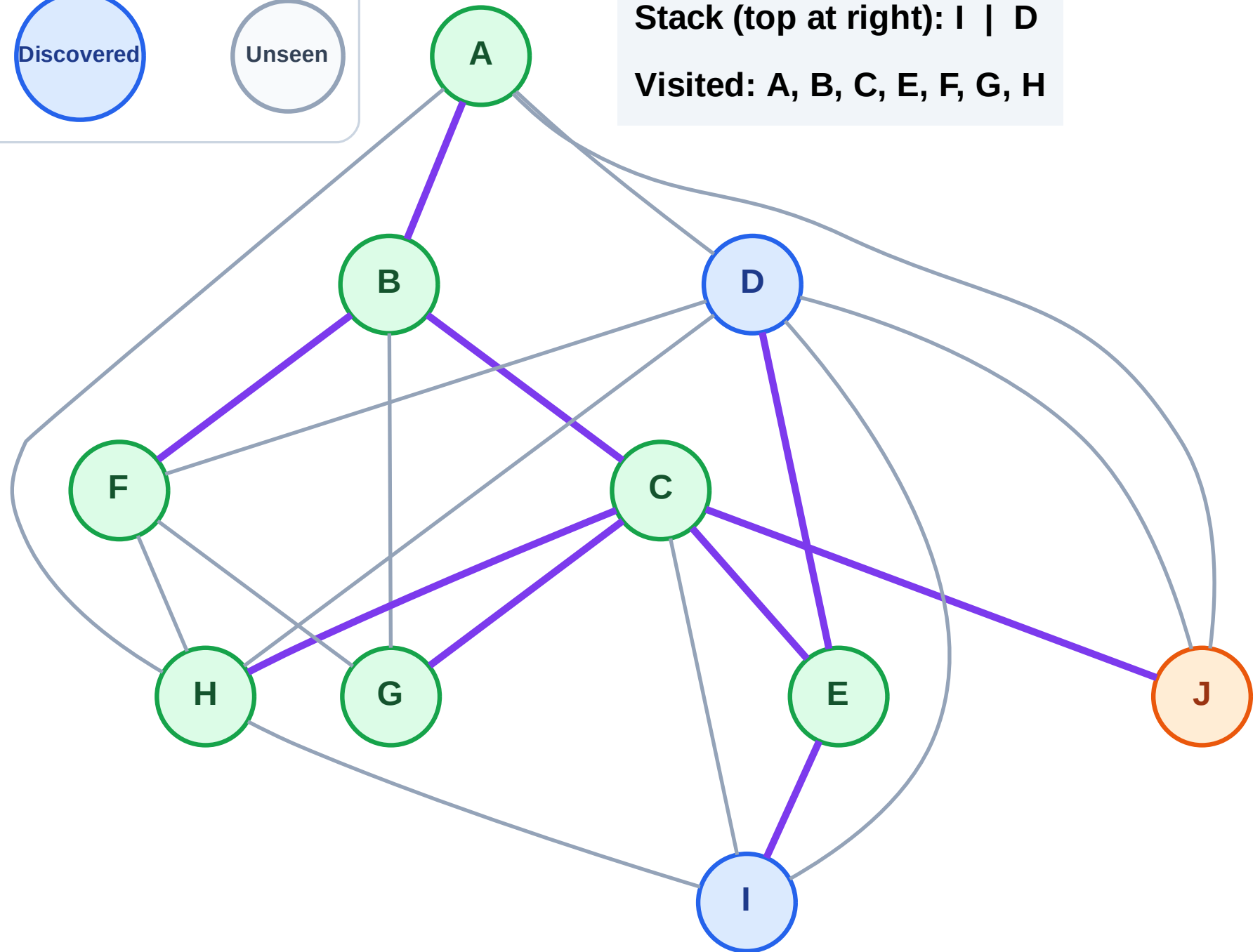
Processing

Discovered

Unseen

Stack (top at right): I | D

Visited: A, B, C, E, F, G, H



# DFS Traversal

Step 17: No new neighbors from J

## Legend

Complete

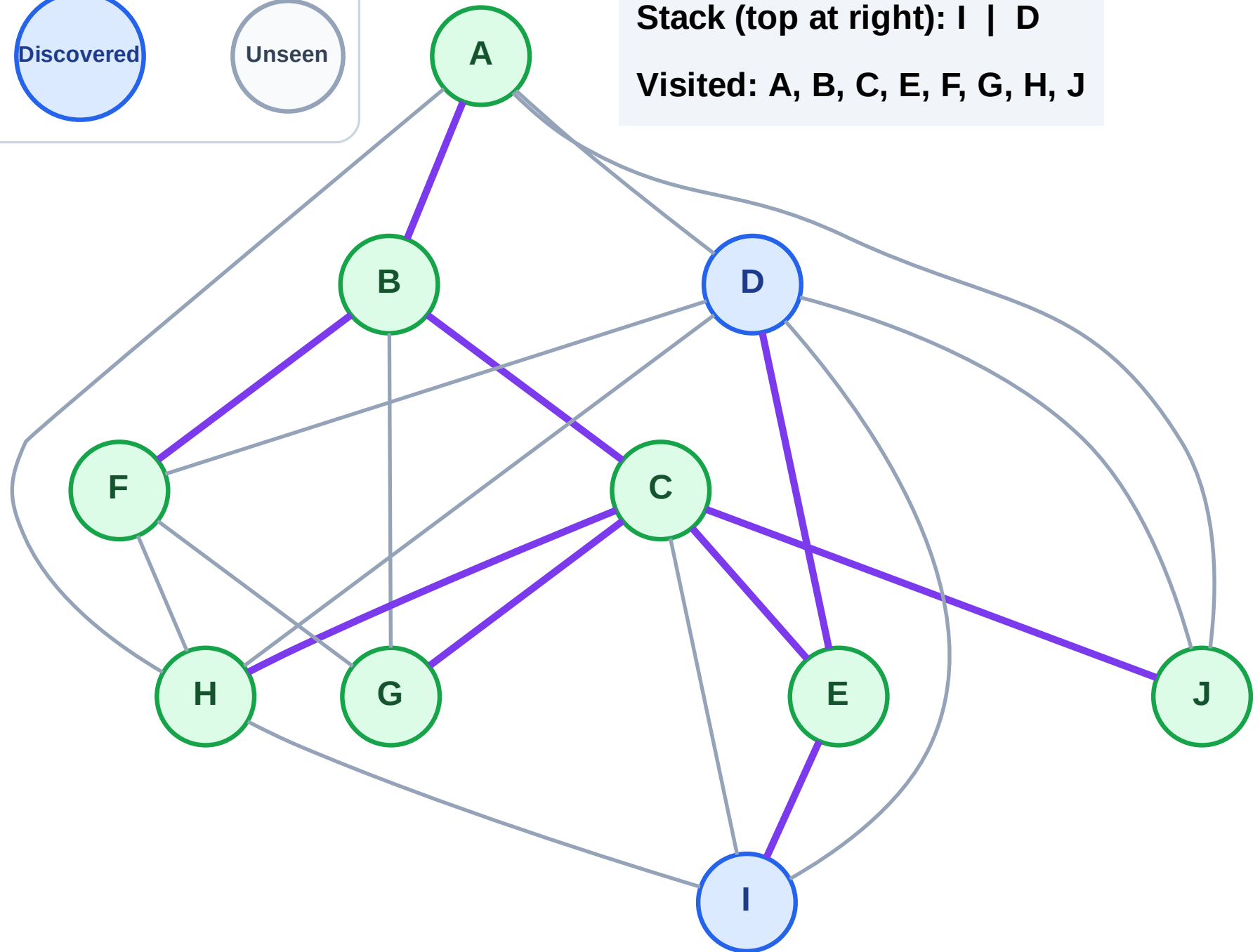
Processing

Discovered

Unseen

Stack (top at right): I | D

Visited: A, B, C, E, F, G, H, J



# DFS Traversal

Step 18: Process node D

## Legend

Complete

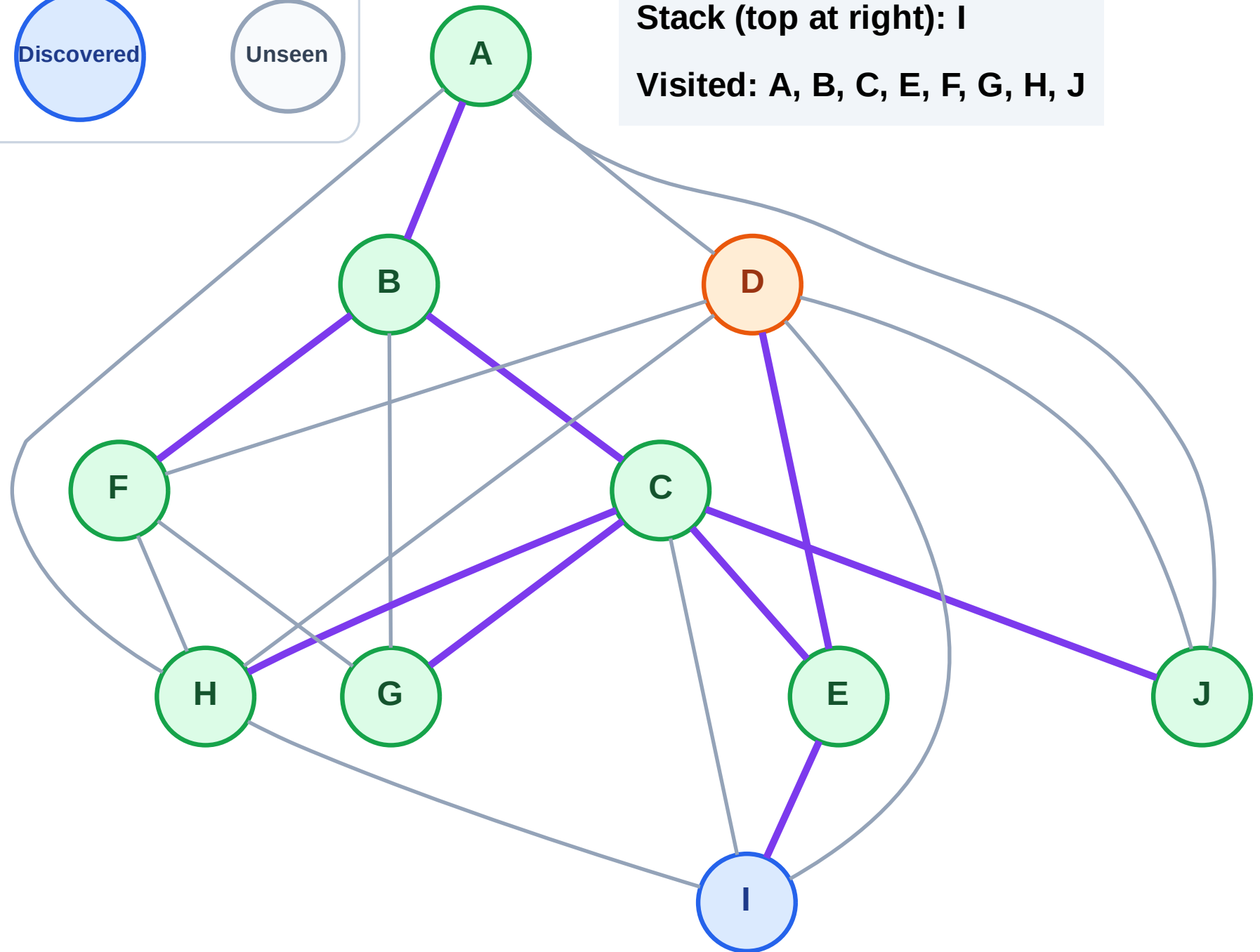
Processing

Discovered

Unseen

Stack (top at right): I

Visited: A, B, C, E, F, G, H, J



# DFS Traversal

Step 19: No new neighbors from D

## Legend

Complete

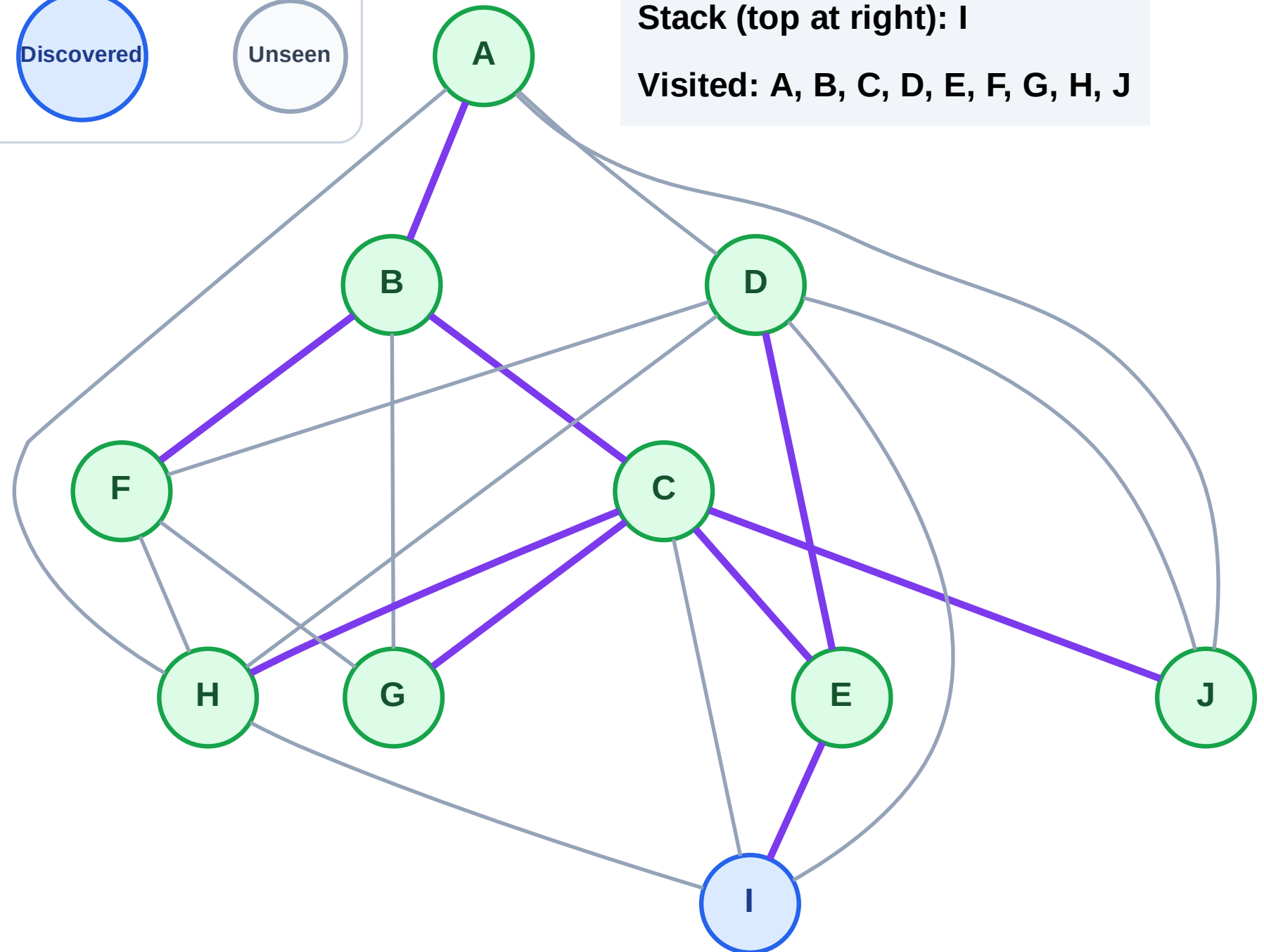
Processing

Discovered

Unseen

Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J

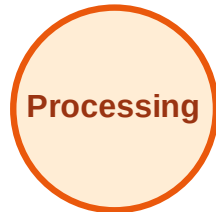




# DFS Traversal

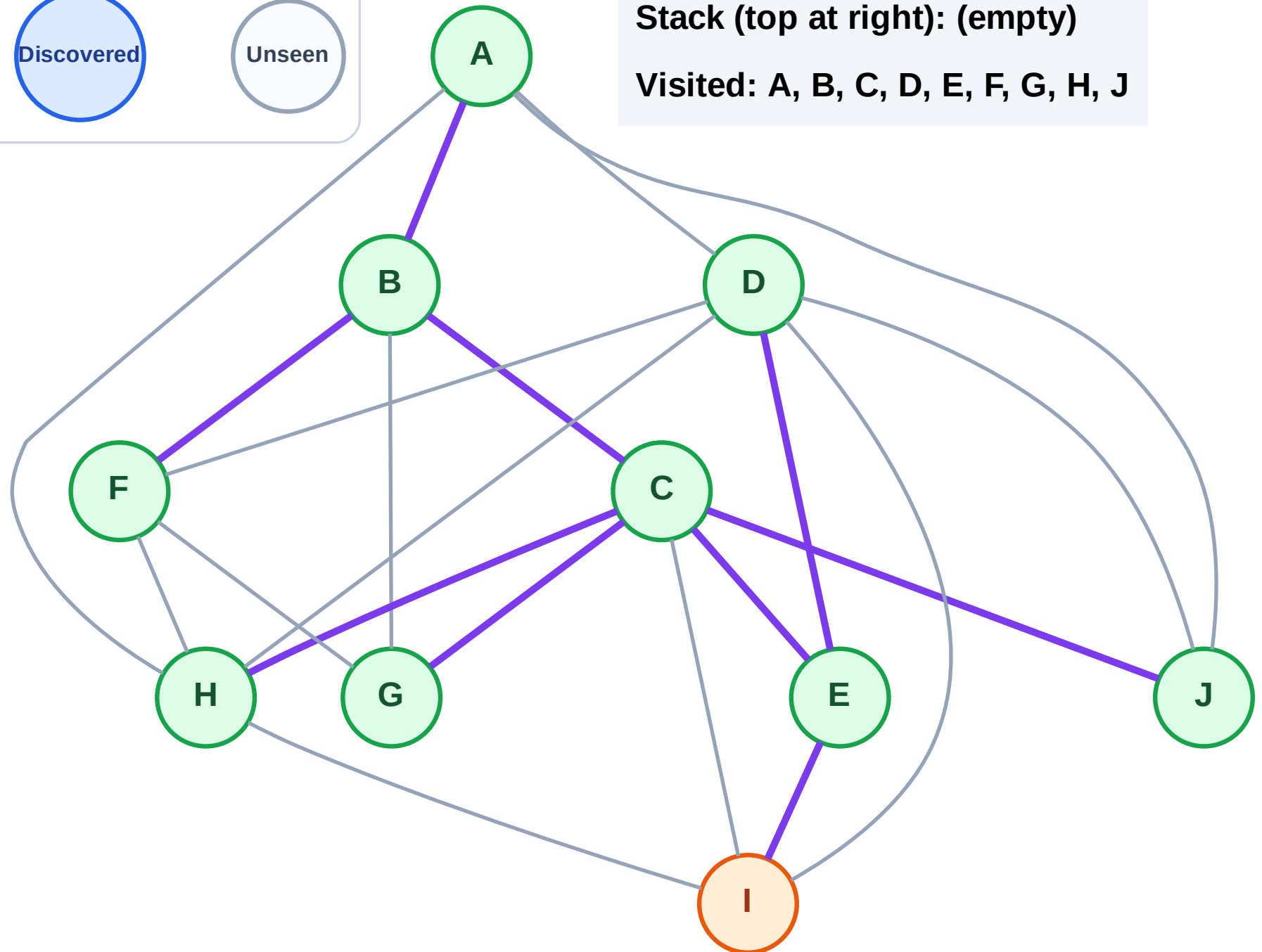
Step 20: Process node I

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



# DFS Traversal

Step 21: No new neighbors from I

## Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

